Cloud Computing (Integration of Database and storage on website)

MUHAMMED RISHAL K

1. Introduction

Cloud computing provides computing resources such as virtual machines, databases, networking and storage over the internet.

In this practical, Microsoft Azure is used to deploy a website using an Ubuntu Linux Virtual Machine. Azure Database for MySQL is integrated with the website to store application data, while Azure Blob Storage is used to store files.

This project demonstrates cloud infrastructure deployment, networking, database connectivity, storage integration and website hosting.

2. Objective

- To deploy a website using Microsoft Azure.
- To create and configure Azure networking resources.
- To deploy and configure a Linux Virtual Machine.
- To install and configure Apache Web Server.
- To create an Azure MySQL database.
- To integrate the database with the website.
- To create and configure Azure Blob Storage.
- To upload and retrieve files from cloud storage.
- To implement secure communication between cloud resources.
- To test and troubleshoot the complete application.

3. Selected Cloud Platform

Microsoft Azure – Option A

Microsoft Azure was selected because it provides virtual machines, managed databases, cloud storage and networking services required for this project.

4. Requirements

Hardware:

- Laptop or desktop computer.
- Internet connection.
- Sufficient RAM and storage.

Software:

- Microsoft Azure Portal.
- Windows PowerShell.
- Ubuntu Server 24.04 LTS.
- Apache Web Server.
- MySQL Client.
- PHP.
- Web browser.
- SSH client.

5. Environment and Cloud Resources

The following Azure resources were used.

Resource	Configuration
Resource Group	`rizzshaal`
Region	Central US
Virtual Network	`vnet-rizz`
Subnet	`subnet-web`
Network Security Group	`nsg-rizz-web`
Virtual Machine	`vm-rizz-web`
Operating System	Ubuntu 24.04
Web Server	Apache
Database Server	`mysql-rizz-web`

Database	`website_db`
Storage Account	`rizzwebstorage`
Blob Container	`website-files`

Verify the actual storage account name in Azure Portal before submitting.

6. Implementation Steps

Step 1: Create Resource Group

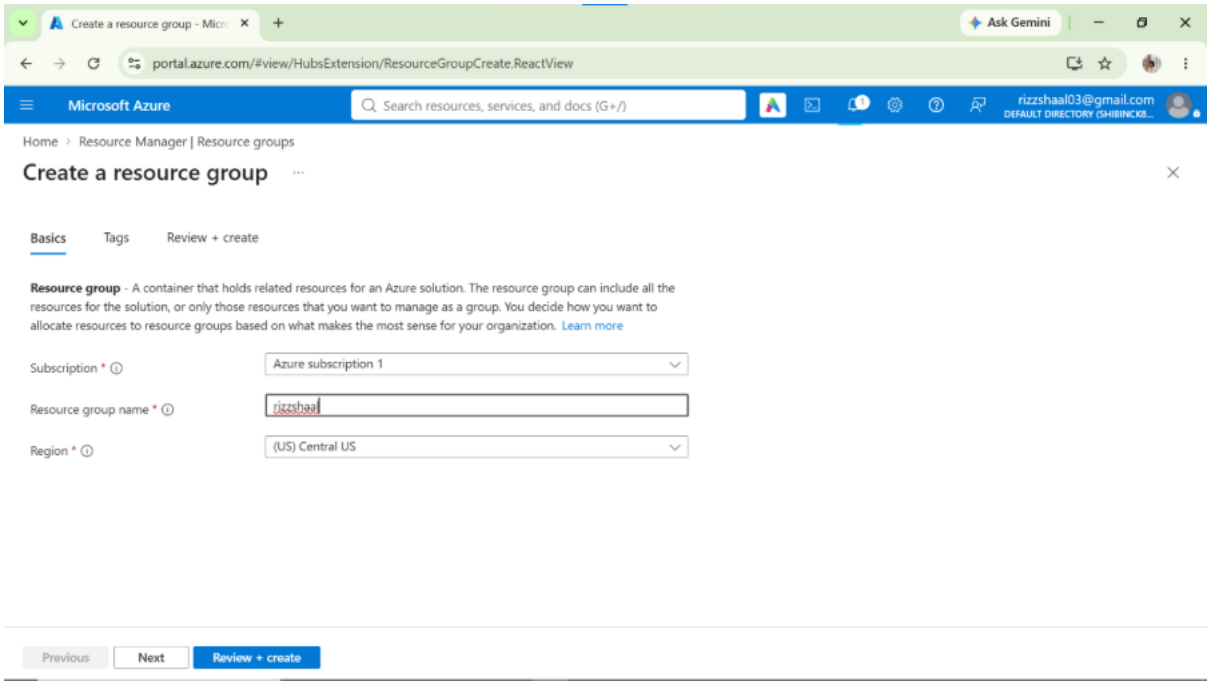
Console path: Azure Portal → Resource groups.

Implementation:

1. Open Microsoft Azure Portal.
2. Search for Resource groups.
3. Open the existing Resource Group `rizzshaal`.
4. Verify the subscription and region.

Explanation:

A Resource Group is a logical container used to organize and manage related Azure resources.



Step 2: Create Virtual Network

Console path: Azure Portal → Virtual networks → Create.

Configuration:

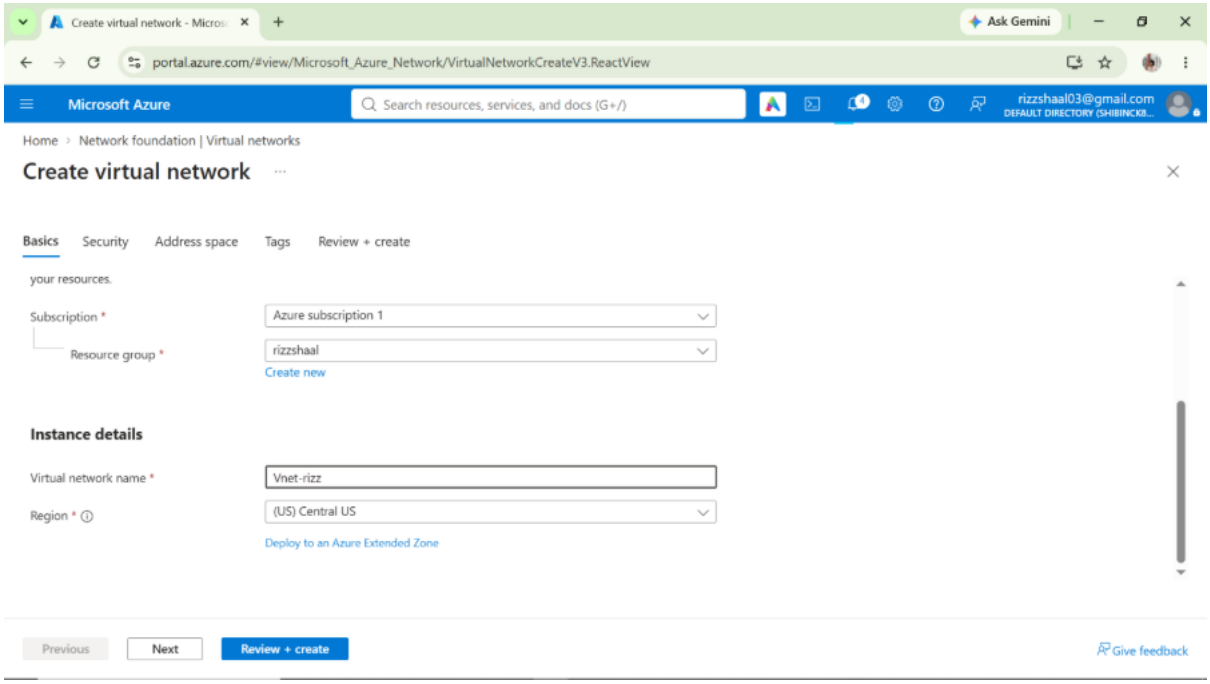
Setting	Value
Resource Group	rizzshaal
Name	vnet-rizz
Region	Central US
Address Space	10.0.0.0/16
Subnet	subnet-web
Subnet Range	10.0.1.0/24

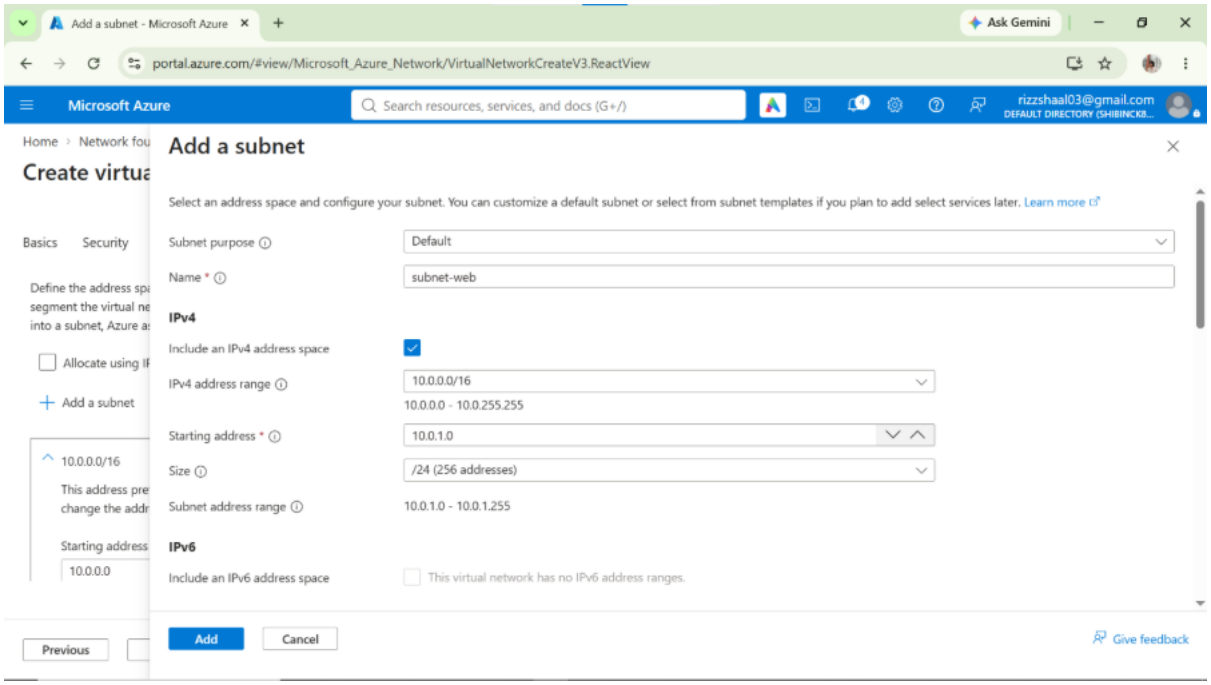
Implementation:

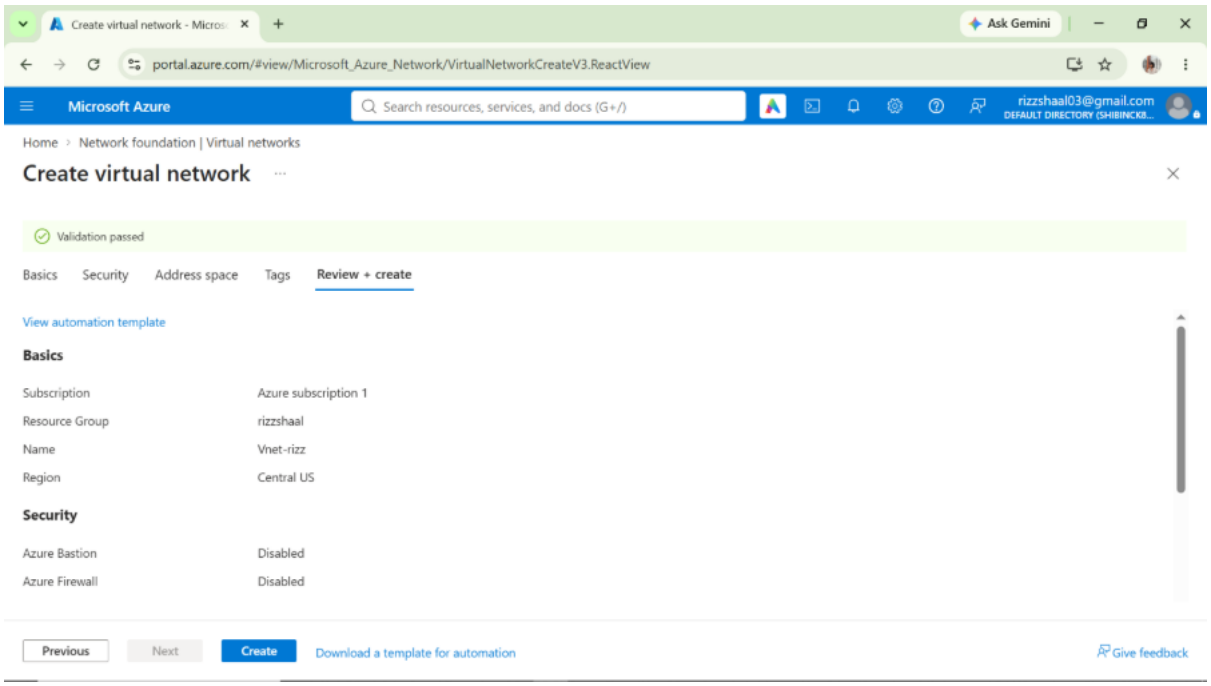
1. Open Virtual Networks.
2. Click Create.
3. Select Resource Group `rizzshaal`.
4. Enter the VNet name.
5. Select Central US.
6. Configure the IPv4 address space.
7. Create the subnet.
8. Review and create.

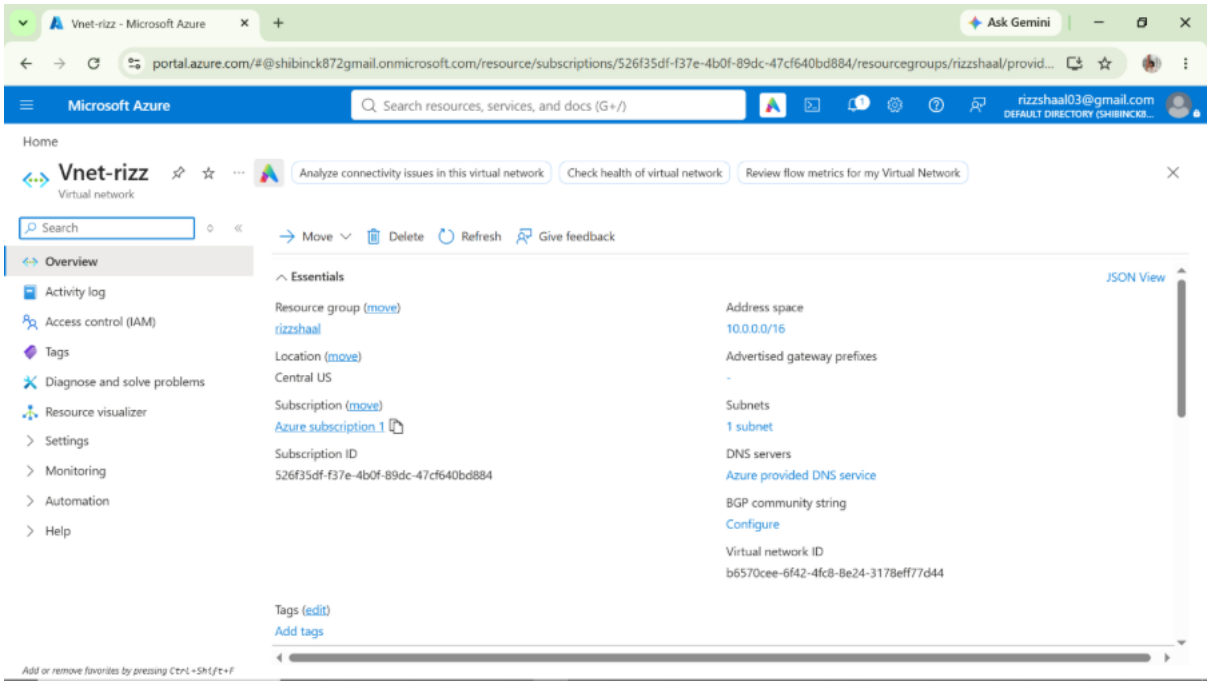
Explanation:

A Virtual Network provides a private network environment for communication between Azure resources.




Add a subnet - Microsoft Azure
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_Network/VirtualNetworkCreateV3.ReactView
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home
Network fou
Create virtua
Add a subnet
Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more
Basics
Security
Subnet purpose
Default
Name *
subnet-web
IPv4
Include an IPv4 address space
IPv4 address range
10.0.0.0/16
10.0.0.0 - 10.0.255.255
Starting address *
10.0.1.0
Size
/24 (256 addresses)
Subnet address range
10.0.1.0 - 10.0.1.255
IPv6
Include an IPv6 address space
This virtual network has no IPv6 address ranges.
Add
Cancel
Previous
Give feedback


Create virtual network - Micros
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_Network/VirtualNetworkCreateV3.ReactView
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home
Network foundation | Virtual networks
Create virtual network
Validation passed
Basics
Security
Address space
Tags
Review + create
View automation template
Basics
Subscription
Azure subscription 1
Resource Group
rizzshaal
Name
Vnet-rizz
Region
Central US
Security
Azure Bastion
Disabled
Azure Firewall
Disabled
Previous
Next
Create
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Step 3: Configure Network Security Group

Console path: Azure Portal → Network Security Groups → `nsg-rizz-web`.

Inbound security rules:

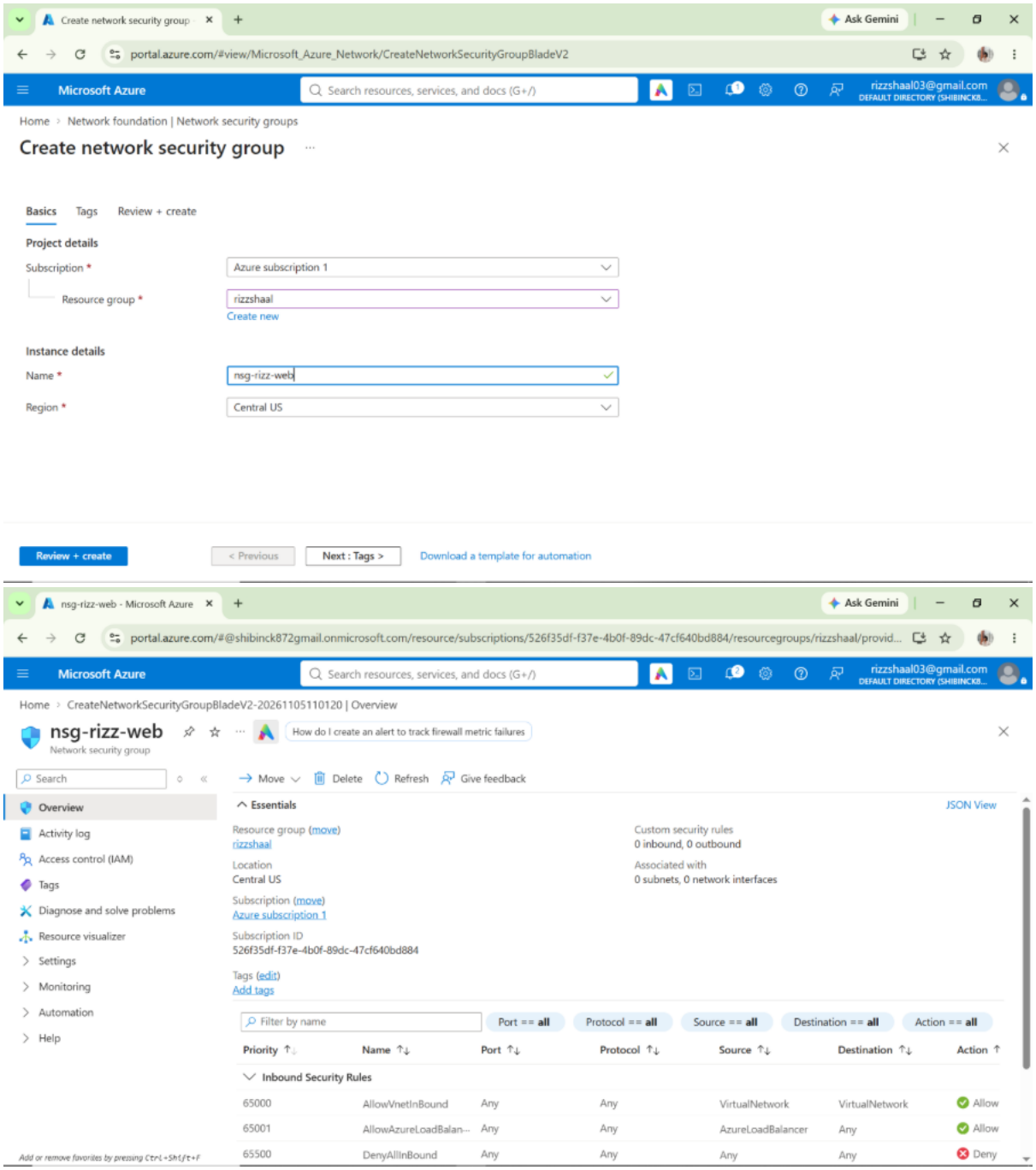
Rule	Destination Port	Priority	Protocol
SSH	22	100	TCP
HTTP	80	110	TCP
HTTPS	443	120	TCP

Implementation:

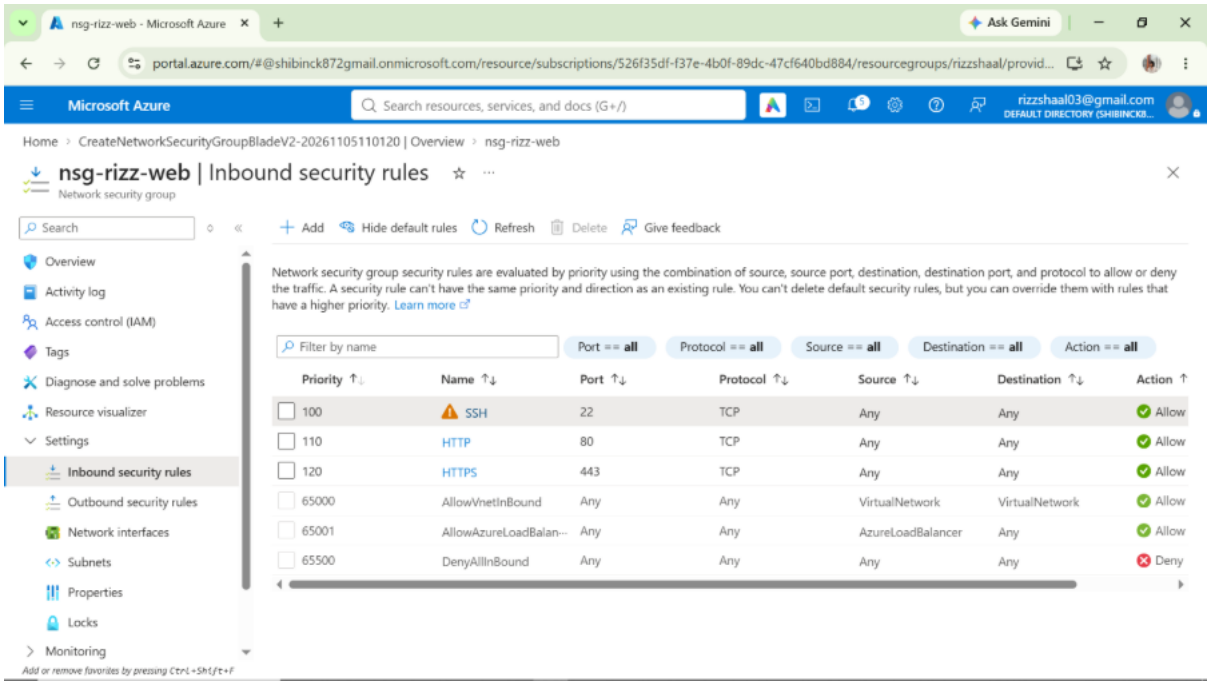
1. Create the NSG.
2. Open Inbound Security Rules.
3. Add SSH, HTTP and HTTPS rules.
4. Configure the destination ports and priorities.
5. Restrict SSH access to the required source IP.
6. Associate the NSG with the VM subnet or network interface.

Explanation:

An NSG controls incoming and outgoing network traffic and protects Azure resources from unauthorized access.


Create network security group
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_Network/CreateNetworkSecurityGroupBladeV2
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Network foundation | Network security groups
Create network security group
Basics
Tags
Review + create
Project details
Subscription *
Azure subscription 1
Resource group *
rizzshaal
Create new
Instance details
Name *
nsg-rizz-web
Region *
Central US
Review + create
< Previous
Next : Tags >
Download a template for automation

nsg-rizz-web - Microsoft Azure
Ask Gemini
portal.azure.com/#@shibinck872gmail.onmicrosoft.com/resource/subscriptions/526f35df-f37e-4b0f-89dc-47cf640bd884/resourcegroups/rizzshaal/provid...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › CreateNetworkSecurityGroupBladeV2-20261105110120 | Overview
nsg-rizz-web
Network security group
How do I create an alert to track firewall metric failures
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Monitoring
Automation
Help
Move
Delete
Refresh
Give feedback
Essentials
JSON View
Resource group (move)
rizzshaal
Location
Central US
Subscription (move)
Azure subscription 1
Subscription ID
526f35df-f37e-4b0f-89dc-47cf640bd884
Tags (edit)
Add tags
Custom security rules
0 inbound, 0 outbound
Associated with
0 subnets, 0 network interfaces
Filter by name
Port == all
Protocol == all
Source == all
Destination == all
Action == all
Priority
Name
Port
Protocol
Source
Destination
Action
Inbound Security Rules
65000 AllowVnetInBound Any Any VirtualNetwork VirtualNetwork Allow
65001 AllowAzureLoadBalan... Any Any AzureLoadBalancer Any Allow
65500 DenyAllInBound Any Any Any Any Deny
Add or remove favorites by pressing Ctrl+Shift+F



Step 4: Create Linux Virtual Machine

Console path: Azure Portal → Virtual Machines → Create → Azure Virtual Machine.

Configuration:

Setting	Value
Resource Group	rizzshaal
VM Name	vm-rizz-web
Region	Central US
Image	Ubuntu Server 24.04 LTS
Authentication	SSH Public Key
Username	azureuser

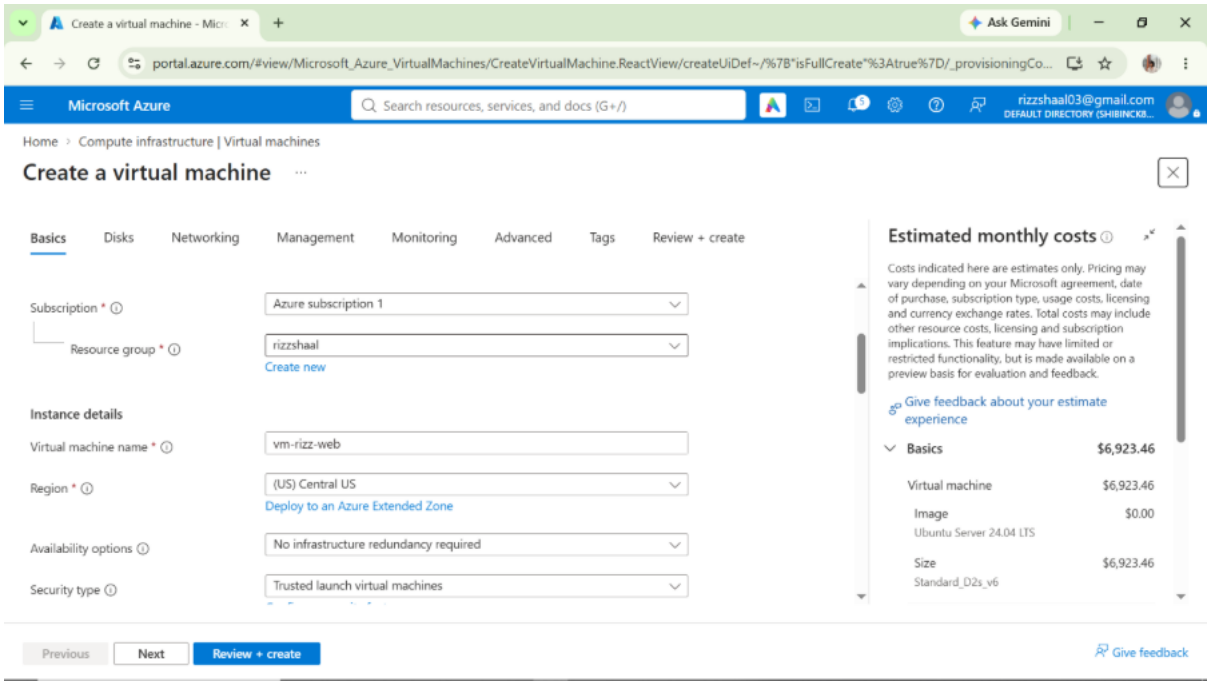
Disk	Standard SSD
VNet	vnet-rizz
Subnet	subnet-web
NSG	nsg-rizz-web

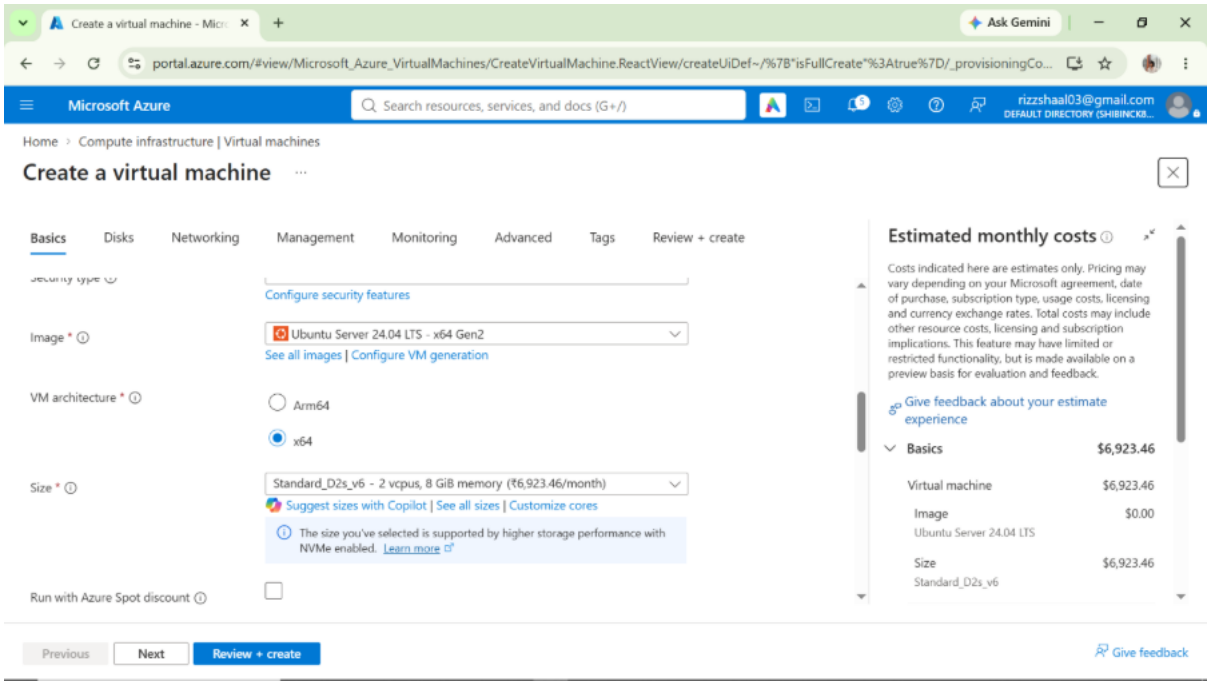
Implementation:

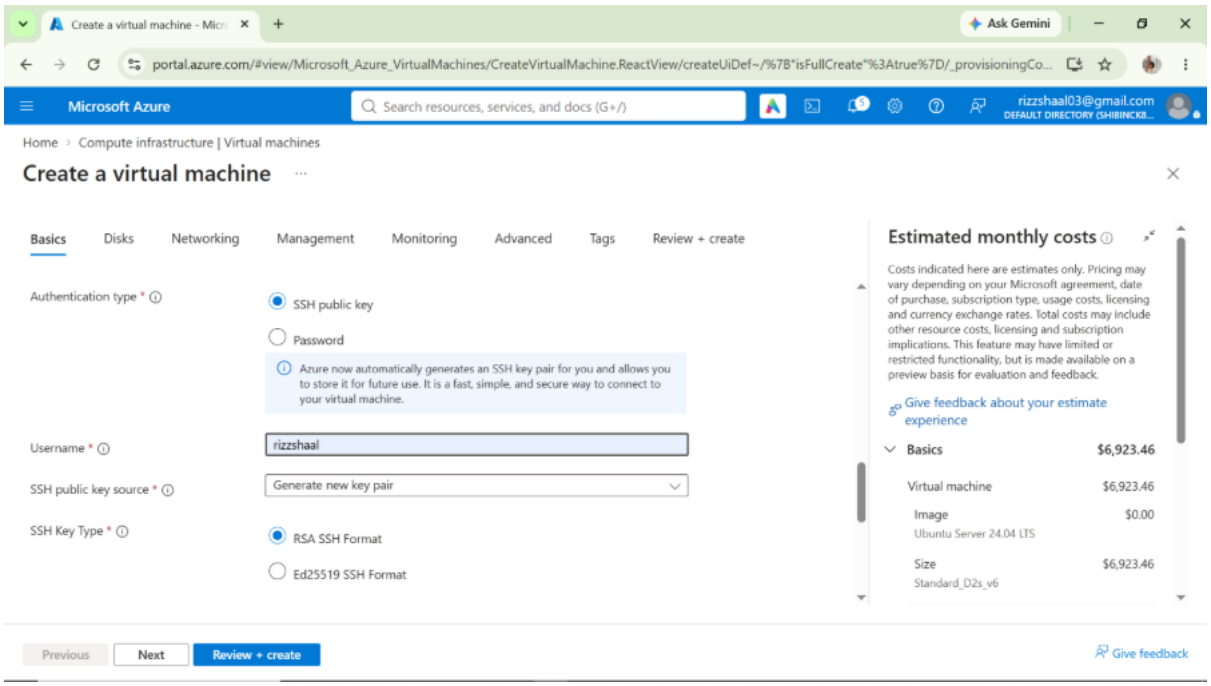
1. Open Virtual Machines.
2. Click Create.
3. Configure the VM details.
4. Select Ubuntu Server 24.04.
5. Configure SSH authentication.
6. Download and save the private key.
7. Configure the operating system disk.
8. Select the existing VNet and subnet.
9. Associate the NSG.
10. Create the VM.
11. Wait for deployment to complete.

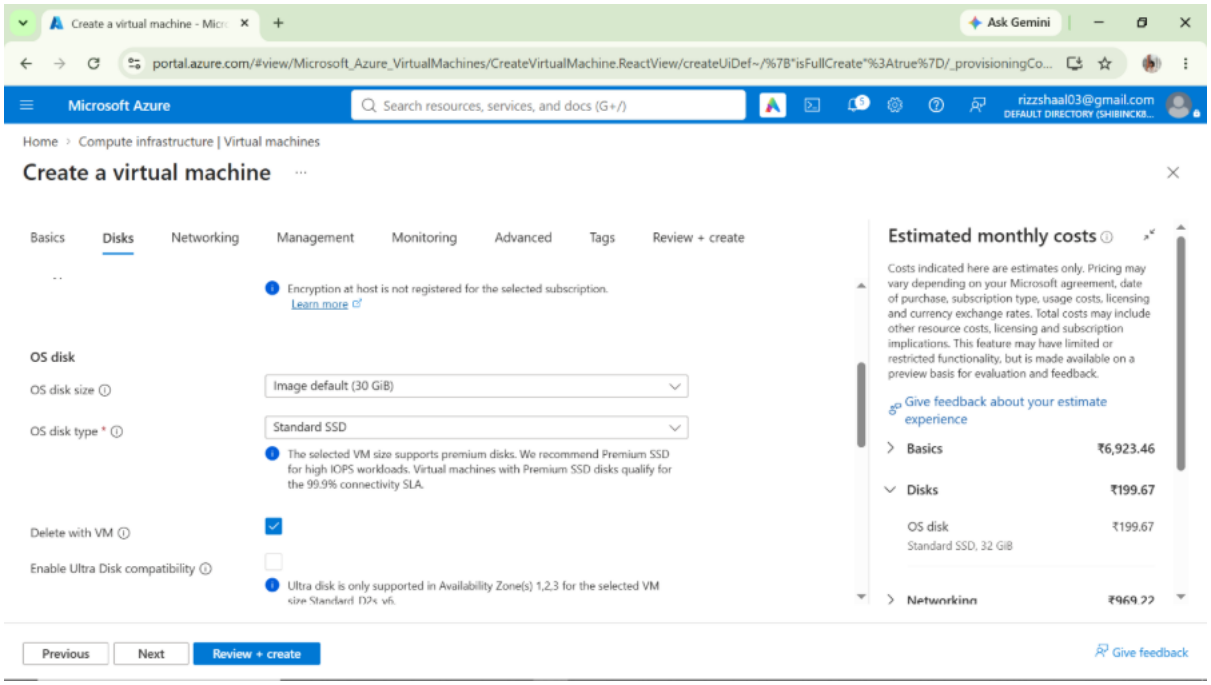
Explanation:

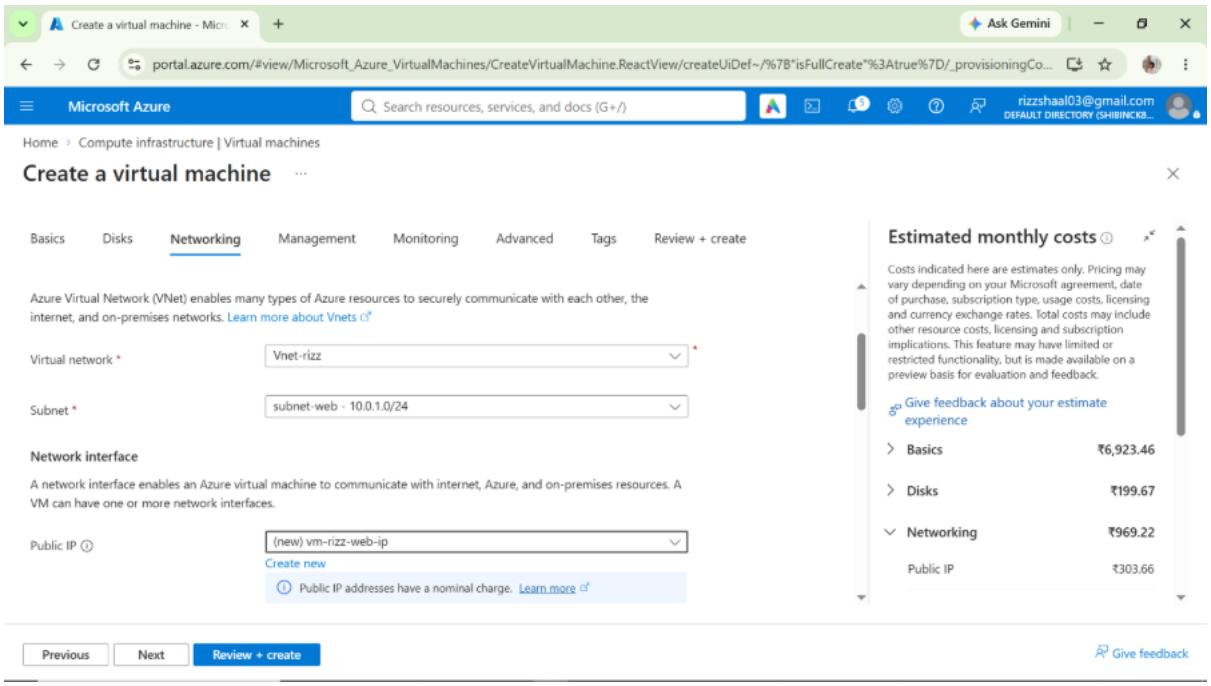
The Virtual Machine provides the computing environment required to host the website and communicate with Azure Database and Storage.

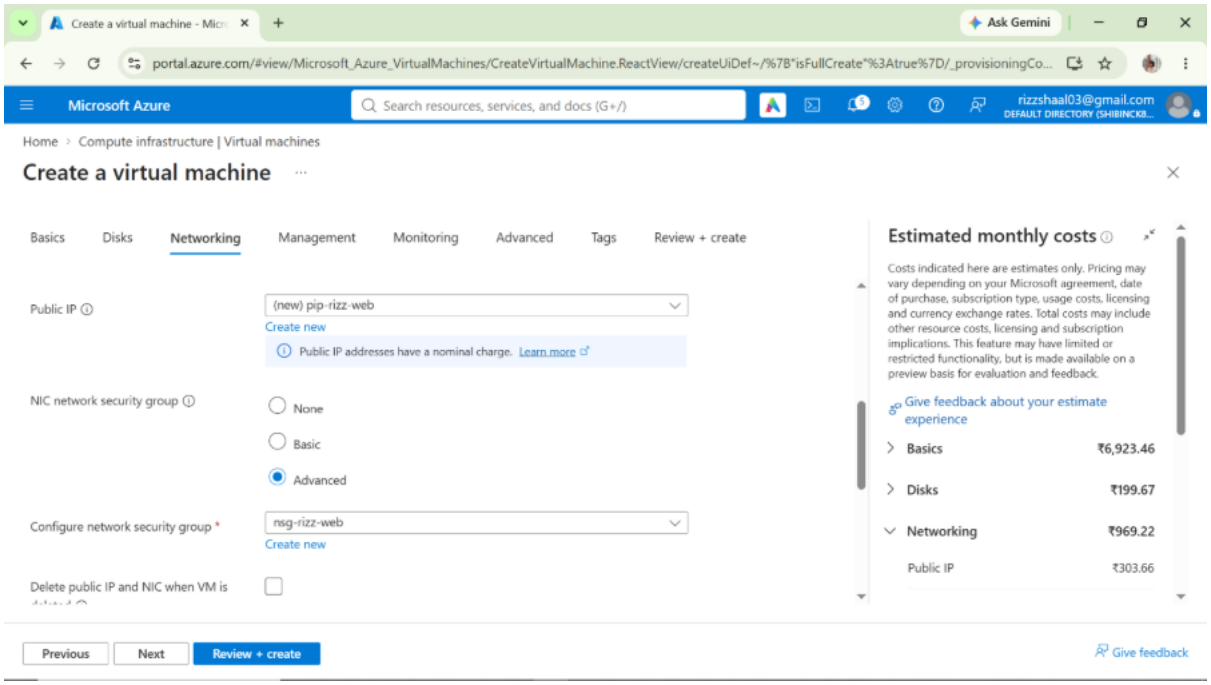

Create a virtual machine - Micr
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_VirtualMachines/CreateVirtualMachine.ReactView/createUiDef~/%7B"isFullCreate"%3Atrue%7D/_provisioningCo...
Microsoft Azure
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DEFAULT DIRECTORY (SHIBINCK8...
Home > Compute infrastructure | Virtual machines
Create a virtual machine
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Subscription *
Azure subscription 1
Resource group *
rizzshaal
Create new
Instance details
Virtual machine name *
vm-rizz-web
Region *
(US) Central US
Deploy to an Azure Extended Zone
Availability options
No infrastructure redundancy required
Security type
Trusted launch virtual machines
Estimated monthly costs
Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.
Give feedback about your estimate experience
Basics
$6,923.46
Virtual machine
$6,923.46
Image
$0.00
Ubuntu Server 24.04 LTS
Size
$6,923.46
Standard_D2s_v6
Previous
Next
Review + create
Give feedback

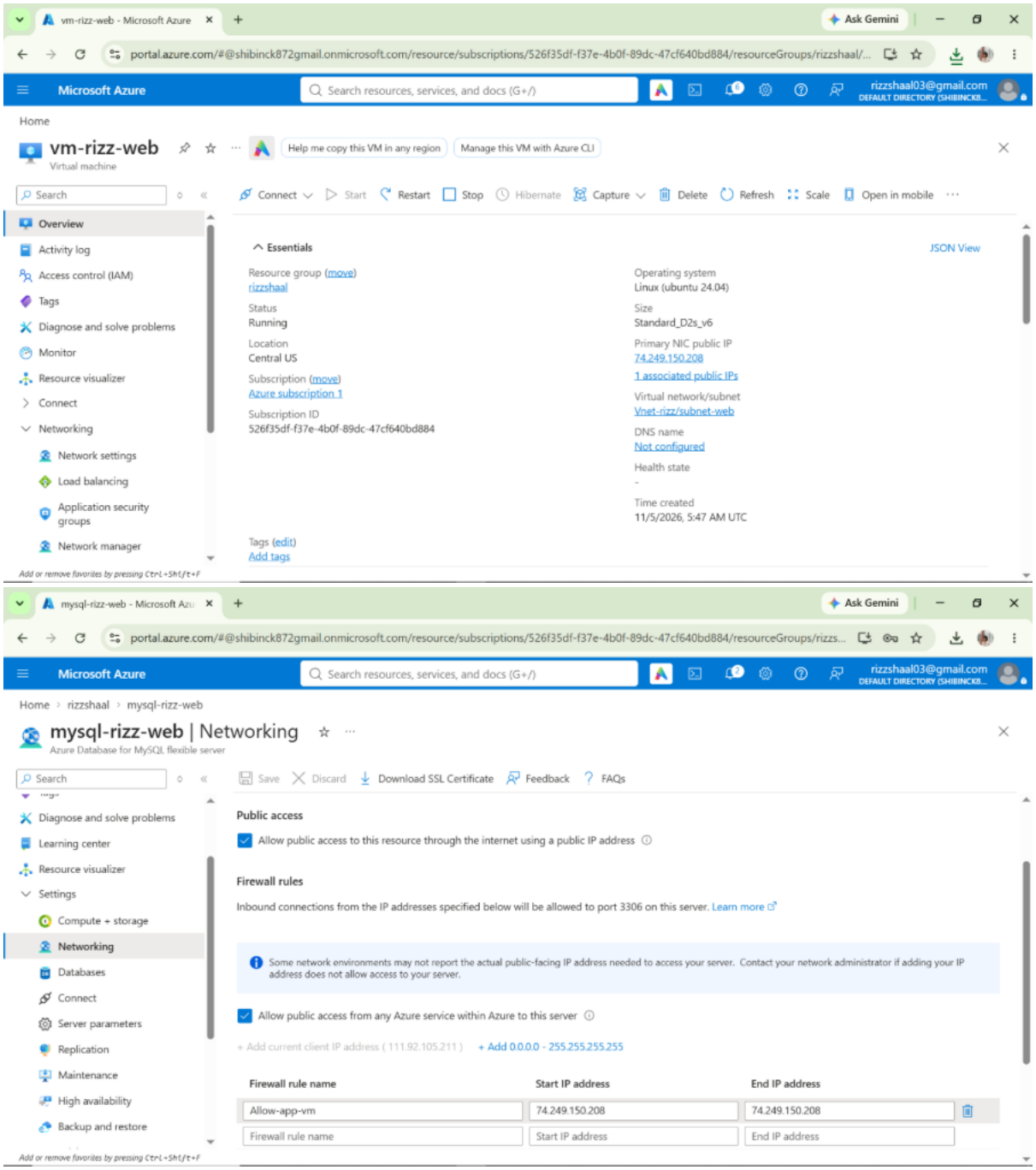

Create a virtual machine - Micr
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_VirtualMachines/CreateVirtualMachine.ReactView/createUiDef~/%7B"isFullCreate"%3Atrue%7D/_provisioningCo...
Microsoft Azure
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DEFAULT DIRECTORY (SHIBINCK8...
Home > Compute infrastructure | Virtual machines
Create a virtual machine
Basics Disks Networking Management Monitoring Advanced Tags Review + create
Configure security features
Image * Ubuntu Server 24.04 LTS - x64 Gen2
See all images | Configure VM generation
VM architecture * Arm64 x64
Size * Standard_D2s_v6 - 2 vcpus, 8 GiB memory (₹6,923.46/month)
Suggest sizes with Copilot | See all sizes | Customize cores
The size you've selected is supported by higher storage performance with NVMe enabled. Learn more
Run with Azure Spot discount
Estimated monthly costs
Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.
Give feedback about your estimate experience
Basics $6,923.46
Virtual machine $6,923.46
Image $0.00
Ubuntu Server 24.04 LTS
Size $6,923.46
Standard_D2s_v6
Previous Next Review + create
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Create a virtual machine - Micr
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_VirtualMachines/CreateVirtualMachine.ReactView/createUiDef~/%7B"isFullCreate"%3Atrue%7D/_provisioningCo...
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Home > Compute infrastructure | Virtual machines
Create a virtual machine
Basics Disks Networking Management Monitoring Advanced Tags Review + create
Authentication type * SSH public key Password
Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.
Username * rizzshaal
SSH public key source * Generate new key pair
SSH Key Type * RSA SSH Format Ed25519 SSH Format
Estimated monthly costs
Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.
Give feedback about your estimate experience
Basics $6,923.46
Virtual machine $6,923.46
Image $0.00
Ubuntu Server 24.04 LTS
Size $6,923.46
Standard_D2s_v6
Previous Next Review + create
Give feedback


Create a virtual machine - Micr
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_VirtualMachines/CreateVirtualMachine.ReactView/createUiDef~/%7B"isFullCreate"%3Atrue%7D/_provisioningCo...
Microsoft Azure
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DEFAULT DIRECTORY (SHIBINCK8...
Home › Compute infrastructure | Virtual machines
Create a virtual machine
Basics Disks Networking Management Monitoring Advanced Tags Review + create
Encryption at host is not registered for the selected subscription.
Learn more
OS disk
OS disk size
Image default (30 GiB)
OS disk type *
Standard SSD
The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.
Delete with VM
Enable Ultra Disk compatibility
Ultra disk is only supported in Availability Zone(s) 1,2,3 for the selected VM size Standard D2s v6.
Estimated monthly costs
Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.
Give feedback about your estimate experience
Basics ₹6,923.46
Disks ₹199.67
OS disk ₹199.67
Standard SSD, 32 GiB
Networking ₹969.22
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portal.azure.com/#view/Microsoft_Azure_VirtualMachines/CreateVirtualMachine.ReactView/createUiDef~/%7B"isFullCreate"%3Atrue%7D/_provisioningCo...
Microsoft Azure
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Create a virtual machine
Basics Disks Networking Management Monitoring Advanced Tags Review + create
Azure Virtual Network (VNet) enables many types of Azure resources to securely communicate with each other, the internet, and on-premises networks. Learn more about Vnets
Virtual network *
Vnet-rizz
Subnet *
subnet-web - 10.0.1.0/24
Network interface
A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.
Public IP
(new) vm-rizz-web-ip
Create new
Public IP addresses have a nominal charge. Learn more
Estimated monthly costs
Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.
Give feedback about your estimate experience
Basics ₹6,923.46
Disks ₹199.67
Networking ₹969.22
Public IP ₹303.66
Previous Next Review + create
Give feedback


Create a virtual machine - Micr
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_VirtualMachines/CreateVirtualMachine.ReactView/createUiDef~/%7B"isFullCreate"%3Atrue%7D/_provisioningCo...
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Create a virtual machine
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Public IP
(new) pip-rizz-web
Create new
Public IP addresses have a nominal charge. Learn more
NIC network security group
None
Basic
Advanced
Configure network security group *
nsg-rizz-web
Create new
Delete public IP and NIC when VM is
Estimated monthly costs
Costs indicated here are estimates only. Pricing may vary depending on your Microsoft agreement, date of purchase, subscription type, usage costs, licensing and currency exchange rates. Total costs may include other resource costs, licensing and subscription implications. This feature may have limited or restricted functionality, but is made available on a preview basis for evaluation and feedback.
Give feedback about your estimate experience
Basics ₹6,923.46
Disks ₹199.67
Networking ₹969.22
Public IP ₹303.66
Previous
Next
Review + create
Give feedback



Step 5: Connect to VM and Install Apache

Console path: Windows → PowerShell.

Connect using SSH:

```
ssh -i "C:\path\to\rizz-web-key.pem" azureuser@YOUR_PUBLIC_IP
```

Replace the path and IP with the actual values.

Update Ubuntu:

```
sudo apt update
```

Install Apache:

```
sudo apt install apache2 -y
```

Start Apache:

```
sudo systemctl enable --now apache2
```

Verify:

```
sudo systemctl status apache2
```

Explanation:

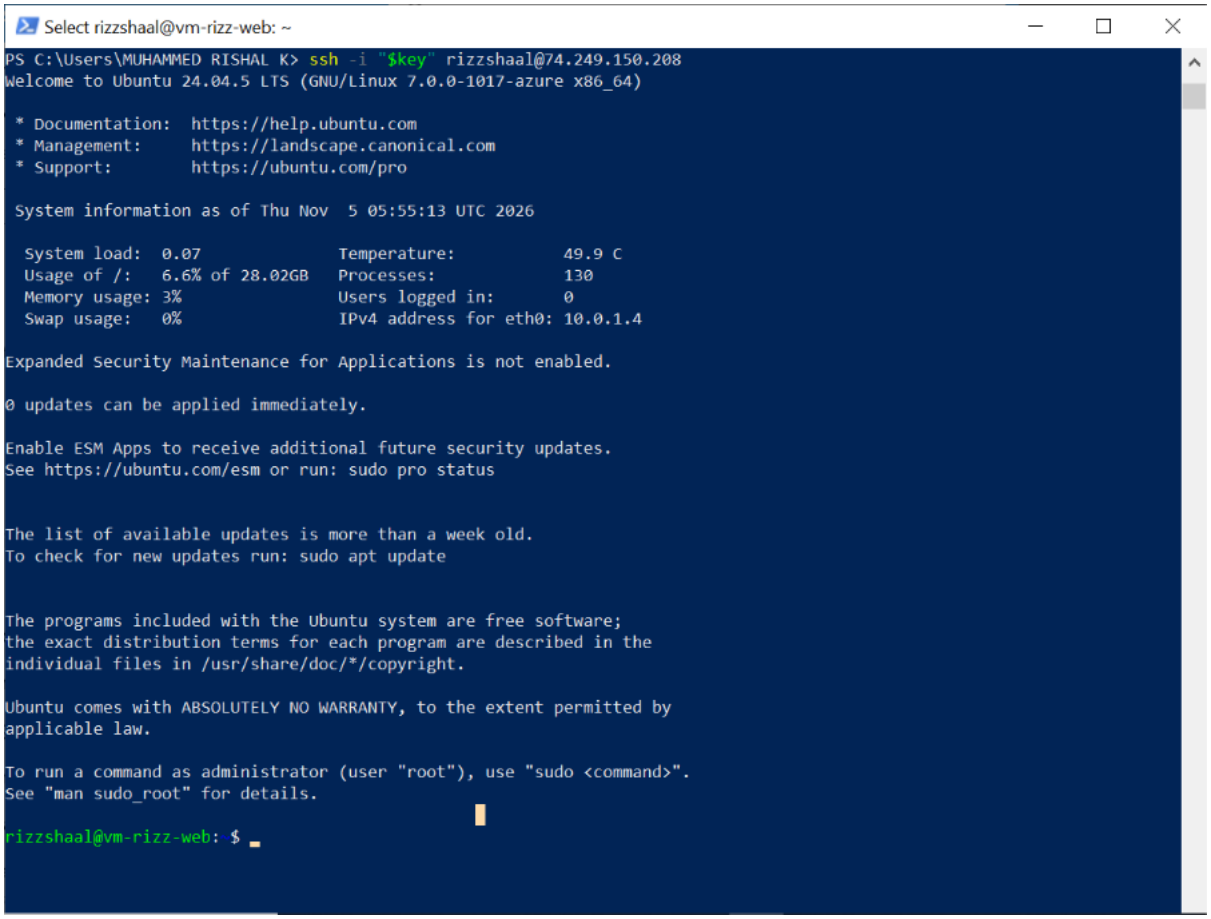
Apache is a web server used to host websites and deliver web pages to users through HTTP and HTTPS.

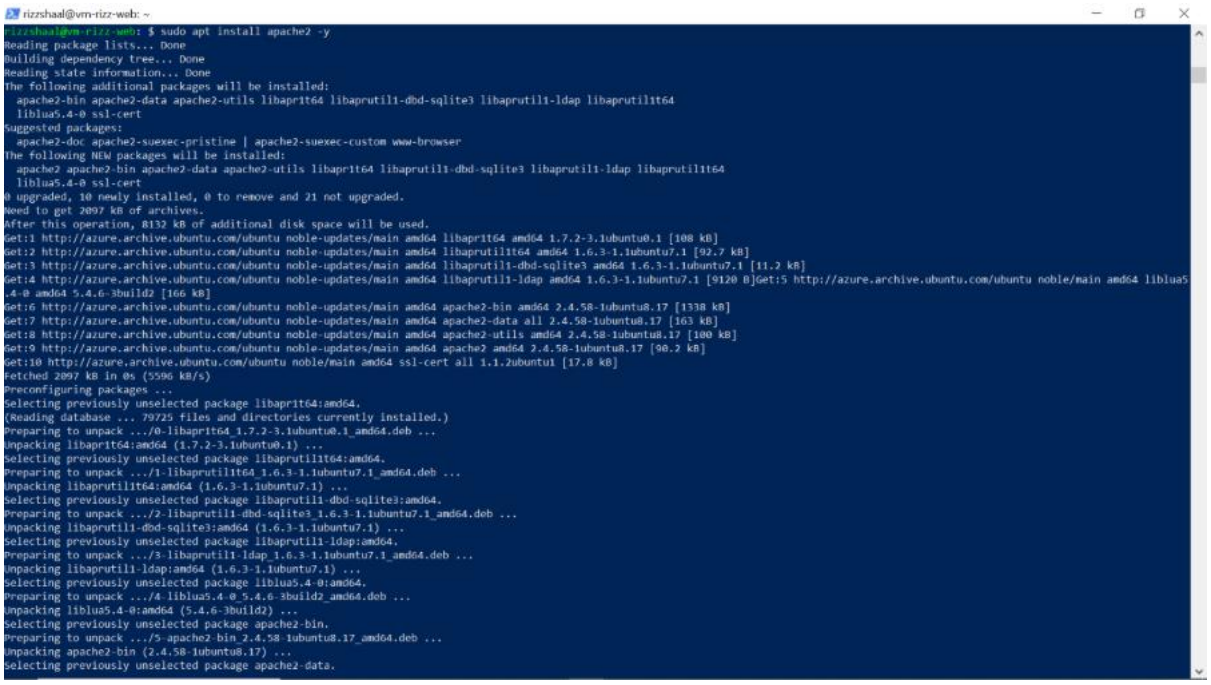
Testing:

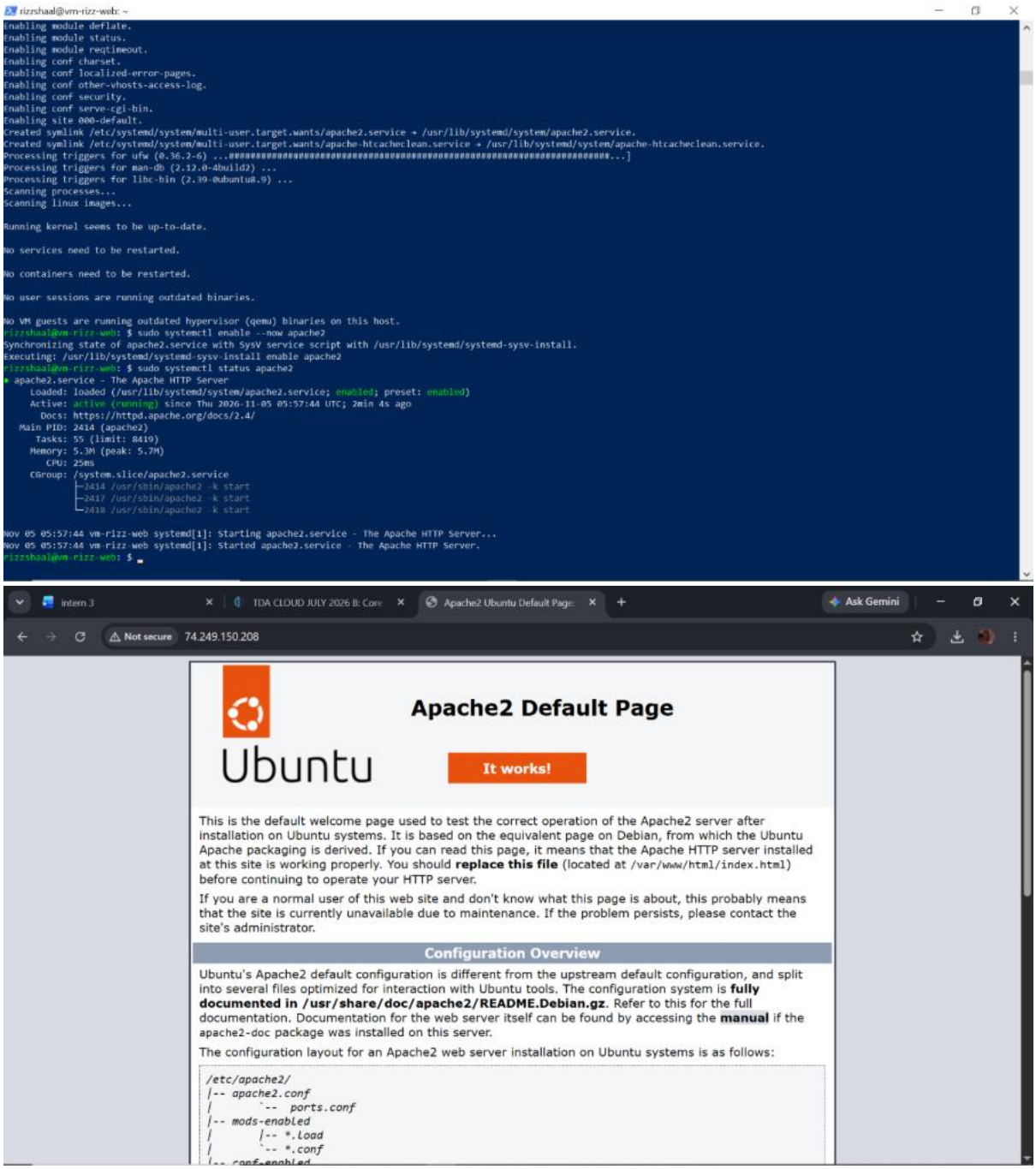
Open a browser and enter:

http://YOUR_PUBLIC_IP

The Apache default webpage should appear.


Select rizzshaal@vm-rizz-web: ~
PS C:\Users\MUHAMMED RISHAL K> ssh -i "$key" rizzshaal@74.249.150.208
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-1017-azure x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Thu Nov 5 05:55:13 UTC 2026
System load: 0.07 Temperature: 49.9 C
Usage of /: 6.6% of 28.02GB Processes: 130
Memory usage: 3% Users logged in: 0
Swap usage: 0% IPv4 address for eth0: 10.0.1.4
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
rizzshaal@vm-rizz-web:~$


rizzshaal@vm-rizz-web: ~
rizzshaal@vm-rizz-web:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
liblua5.4-0 ssl-cert
Suggested packages:
apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 21 not upgraded.
Need to get 2097 kB of archives.
After this operation, 8132 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7.1 [92.7 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7.1 [11.2 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7.1 [9120 B]Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 liblua5
4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.17 [1338 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.17 [163 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.17 [100 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.17 [90.2 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2097 kB in 0s (5596 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 79725 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-1.1ubuntu7.1_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7.1) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7.1_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7.1) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../3-libaprutil1-ldap_1.6.3-1.1ubuntu7.1_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7.1) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../4-liblua5.4-0_5.4.6-3build2_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.6-3build2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../5-apache2-bin_2.4.58-1ubuntu8.17_amd64.deb ...
Unpacking apache2-bin (2.4.58-1ubuntu8.17) ...
Selecting previously unselected package apache2-data.



Step 6: Create Azure Database for MySQL

Console path: Azure Portal → Azure Database for MySQL Flexible Servers → Create.

Configuration:

Setting	Value

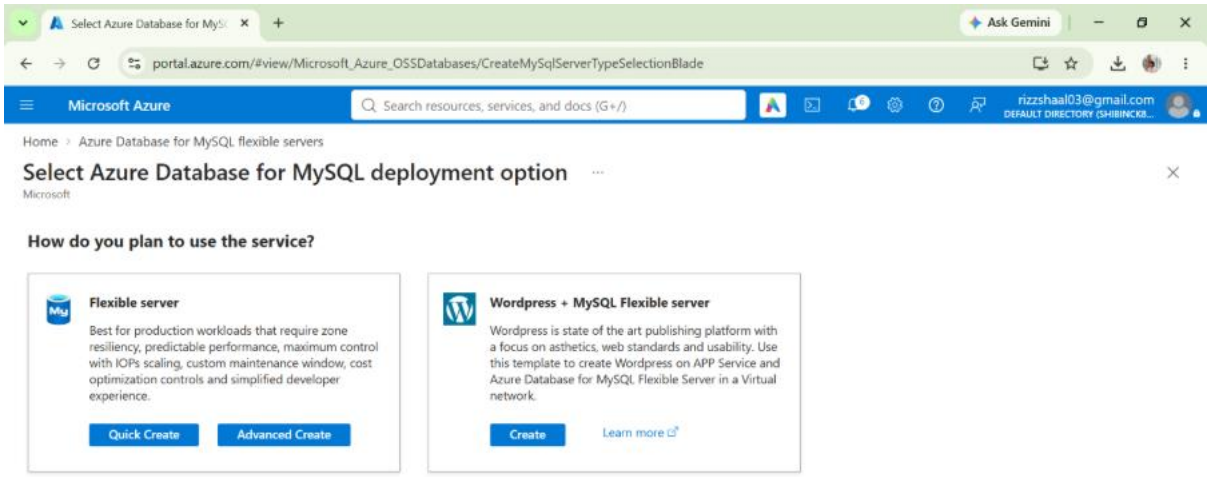
Resource Group	rizzshaal
Server Name	mysql-rizz-web
Region	Central US
Version	MySQL 8.0
Workload	Development
Compute	Burstable
Storage	Minimum suitable size

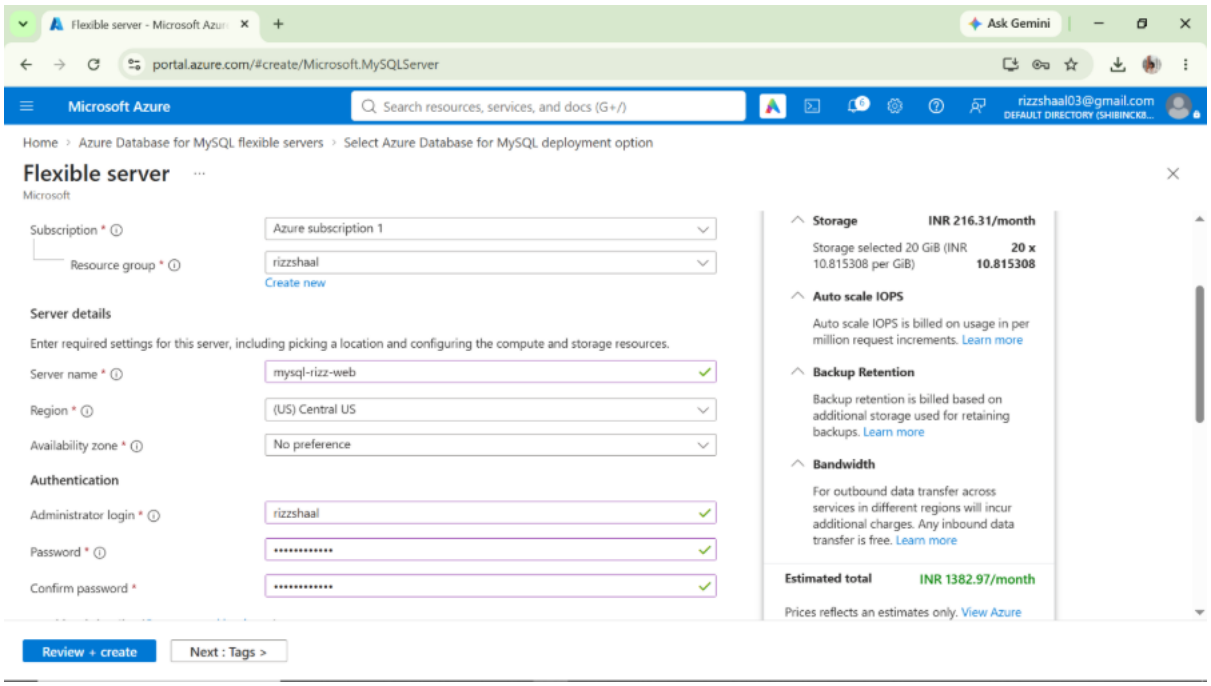
Implementation:

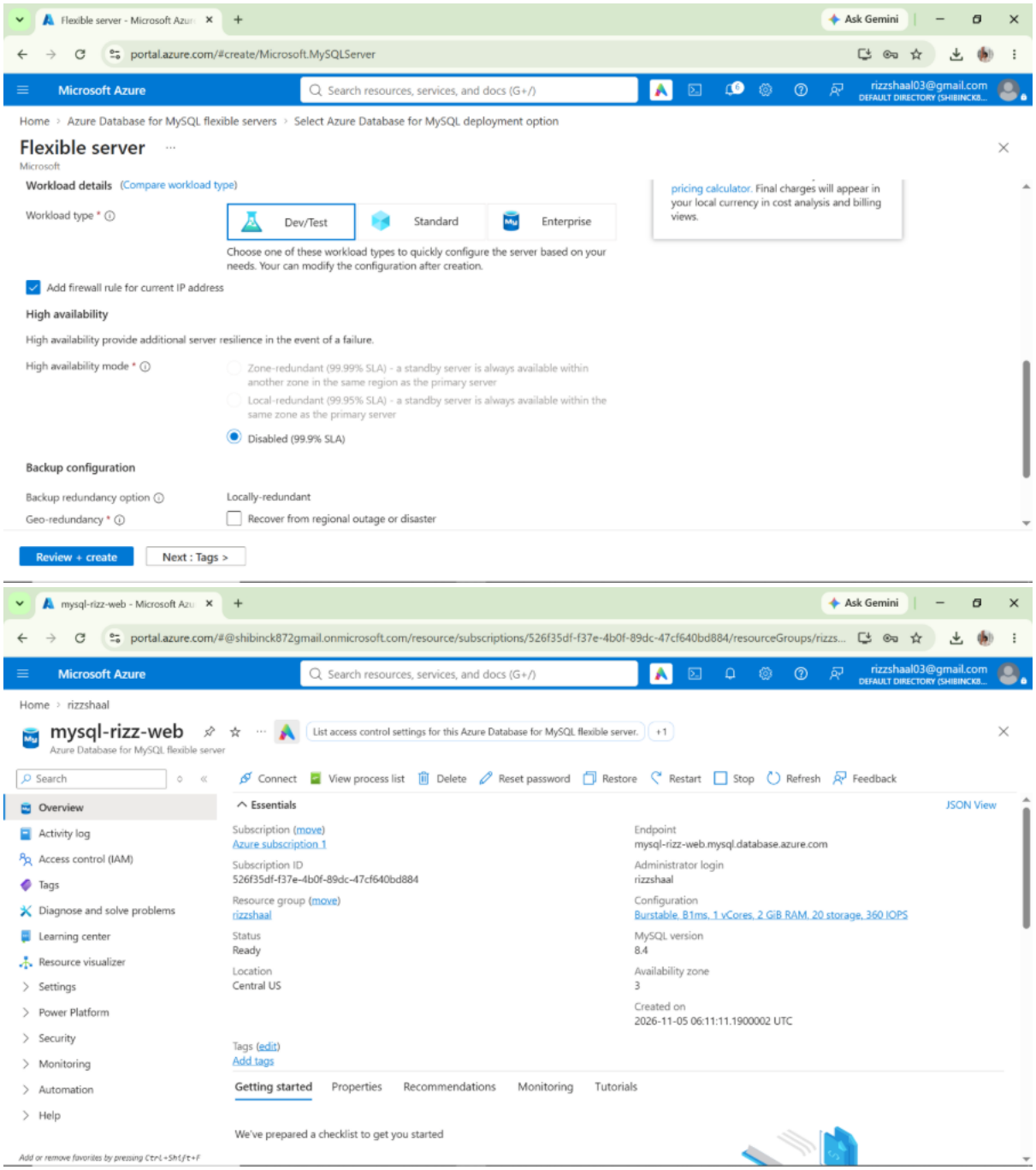
1. Open Azure Database for MySQL.
2. Select Flexible Server.
3. Enter the server name.
4. Select the Resource Group.
5. Select Central US.
6. Configure MySQL version.
7. Configure administrator credentials.
8. Select suitable compute and storage.
9. Configure networking.
10. Review and create.

Explanation:

Azure Database for MySQL is a managed database service used to store application information without manually managing the database server operating system.


Select Azure Database for MyS
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_OSSDatabases/CreateMySqlServerTypeSelectionBlade
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home > Azure Database for MySQL flexible servers
Select Azure Database for MySQL deployment option
Microsoft
How do you plan to use the service?
Flexible server
Best for production workloads that require zone resiliency, predictable performance, maximum control with IOPs scaling, custom maintenance window, cost optimization controls and simplified developer experience.
Quick Create
Advanced Create
Wordpress + MySQL Flexible server
Wordpress is state of the art publishing platform with a focus on asthetics, web standards and usability. Use this template to create Wordpress on APP Service and Azure Database for MySQL Flexible Server in a Virtual network.
Create
Learn more


Flexible server - Microsoft Azur
Ask Gemini
portal.azure.com/#create/Microsoft.MySQLServer
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home > Azure Database for MySQL flexible servers > Select Azure Database for MySQL deployment option
Flexible server
Microsoft
Subscription * Azure subscription 1
Resource group * rizzshaal
Create new
Server details
Enter required settings for this server, including picking a location and configuring the compute and storage resources.
Server name * mysql-rizz-web
Region * (US) Central US
Availability zone * No preference
Authentication
Administrator login * rizzshaal
Password *
Confirm password *
Storage INR 216.31/month
Storage selected 20 GiB (INR 10.815308 per GiB) 20 x 10.815308
Auto scale IOPS
Auto scale IOPS is billed on usage in per million request increments. Learn more
Backup Retention
Backup retention is billed based on additional storage used for retaining backups. Learn more
Bandwidth
For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. Learn more
Estimated total INR 1382.97/month
Prices reflects an estimates only. View Azure
Review + create
Next : Tags >



Step 7: Configure Database Connectivity

Console path: Azure Portal → MySQL Flexible Server → Networking.

Configure firewall access for the VM's public IP.

Connect to the Ubuntu VM using SSH.

Install MySQL Client:

```
sudo apt update
sudo apt install mysql-client -y
```

Test the network connection:

```
nc -vz mysql-rizz-web.mysql.database.azure.com 3306
```

Connect to MySQL:

```
mysql -h mysql-rizz-web.mysql.database.azure.com -u mysqladmin -p --
ssl-mode=REQUIRED
```

Enter the database administrator password.

Create database:

```
CREATE DATABASE website_db;
```

Select database:

```
USE website_db;
```

Create table:

```
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100)
);
```

Insert sample data:

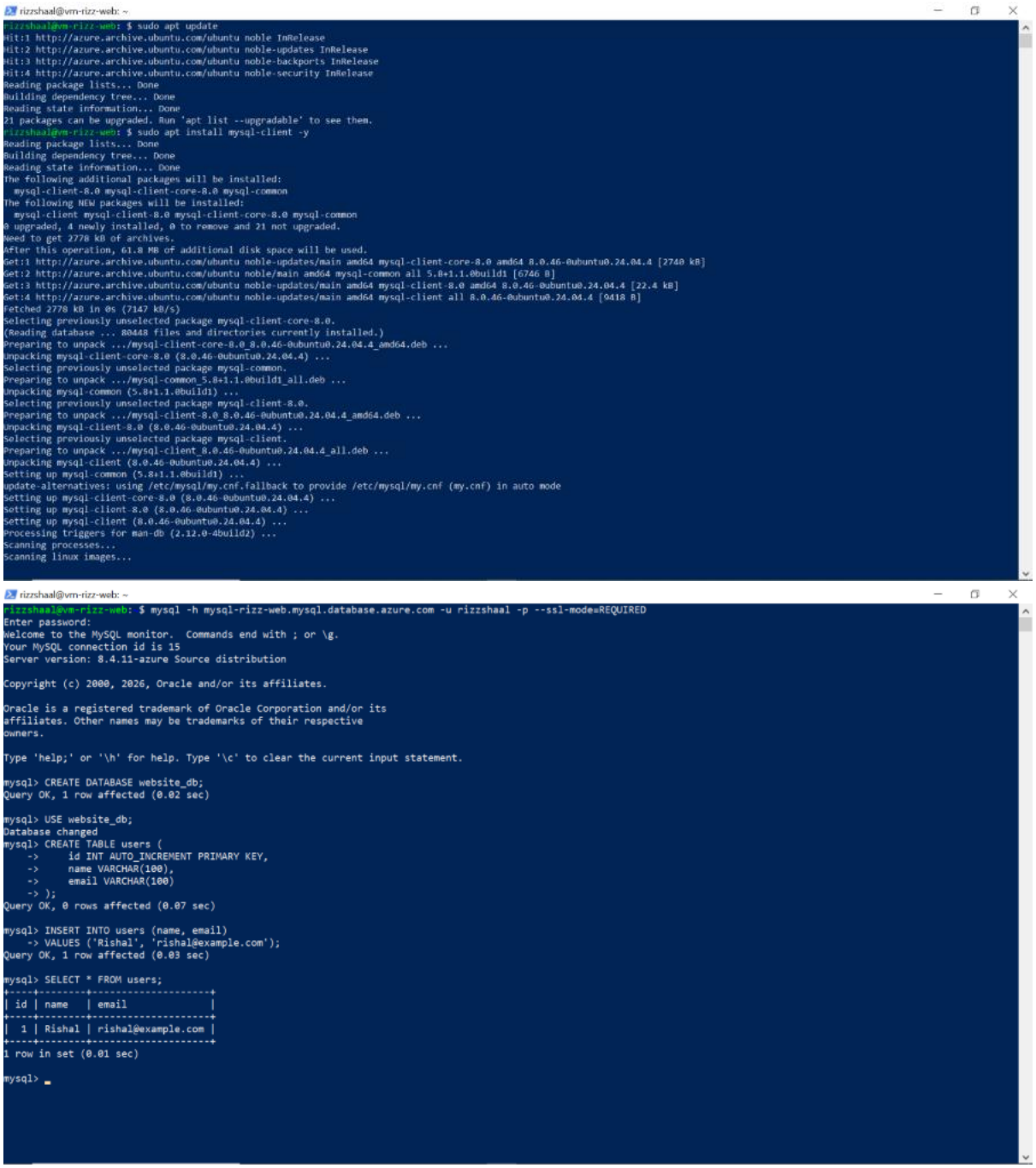
```
INSERT INTO users (name, email)
VALUES ('Rishal', 'rishal@example.com');
```

Verify:

```
SELECT * FROM users;
```

Explanation:

This step establishes communication between the Linux VM and Azure MySQL. The database and table are created to store website user information.



Step 8: Create Azure Storage Account

Console path: Azure Portal → Storage Accounts → Create.

Configuration:

Setting	Value

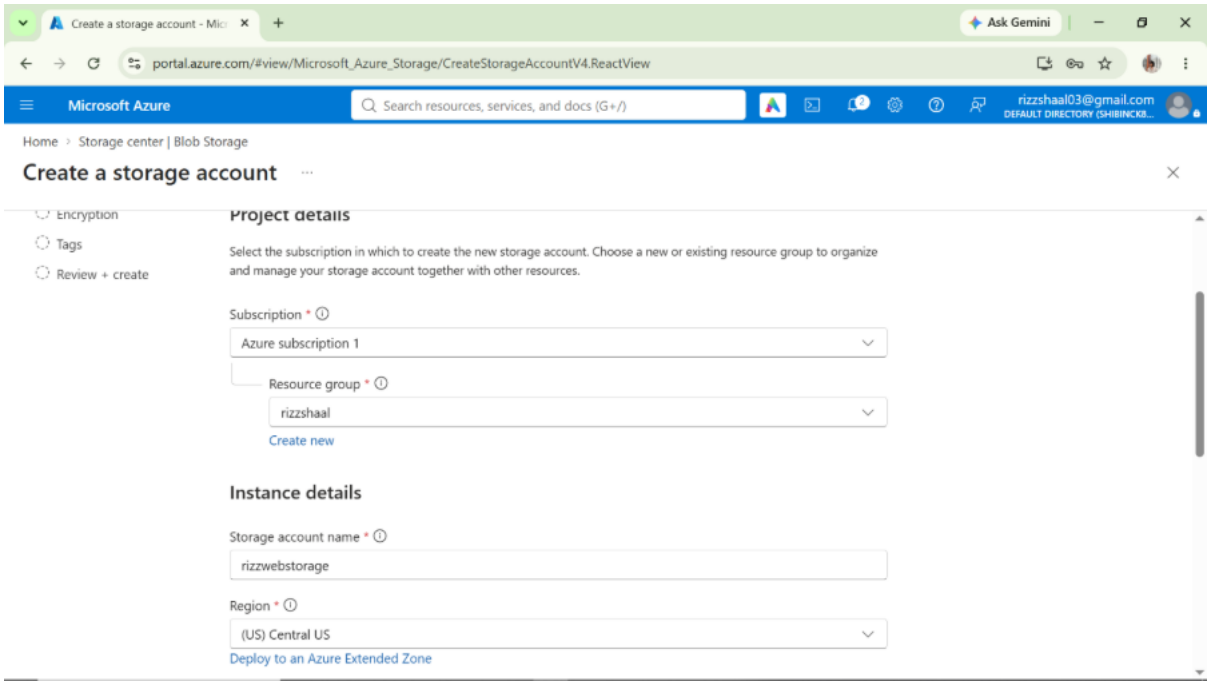
Resource Group	rizzshaal
Storage Name	rizzwebstorage
Region	Central US
Performance	Standard
Redundancy	LRS
Secure Transfer	Enabled
Minimum TLS	1.2

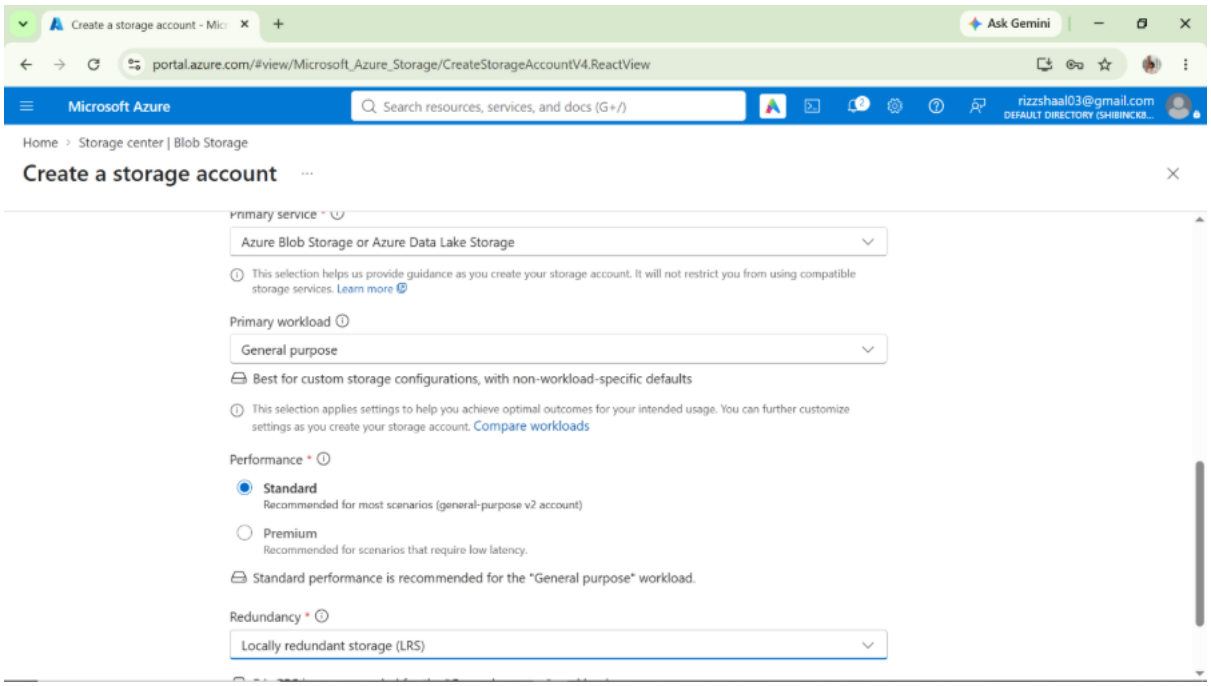
Implementation:

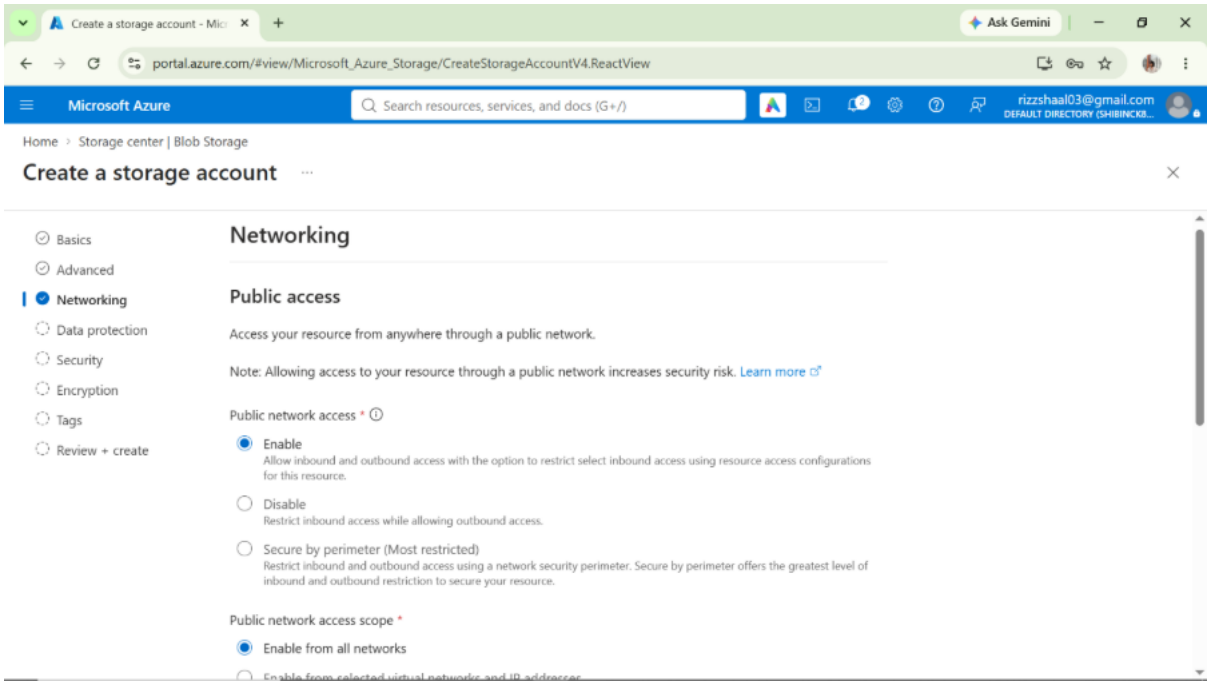
1. Open Storage Accounts.
2. Click Create.
3. Select Resource Group.
4. Enter storage account name.
5. Select Central US.
6. Select Standard performance.
7. Select LRS redundancy.
8. Configure security settings.
9. Review and create.

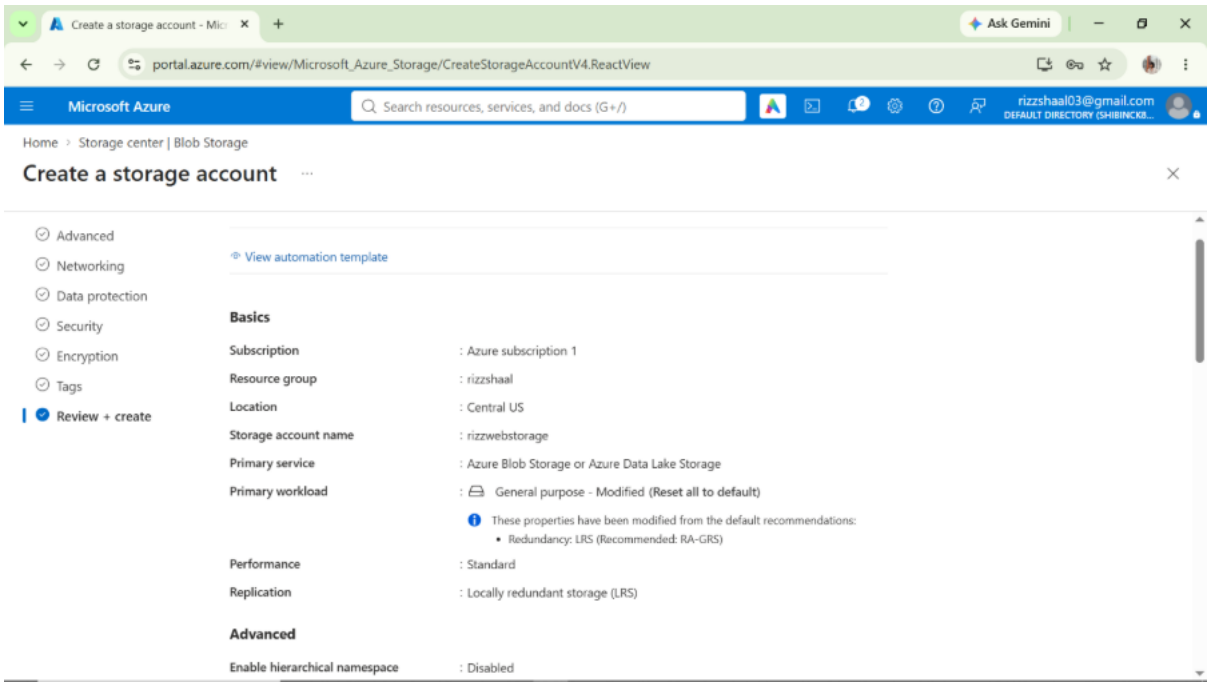
Explanation:

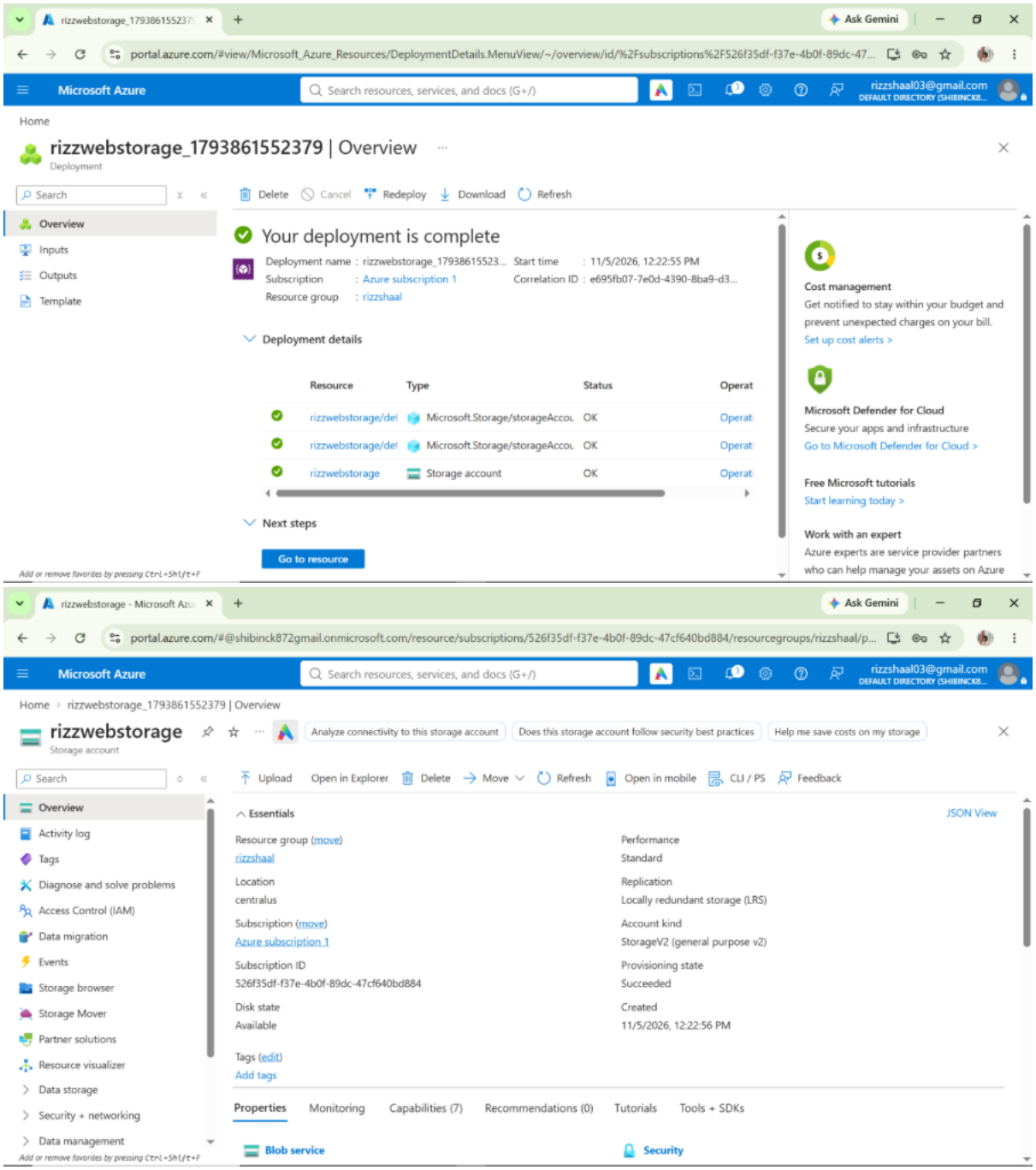
Azure Storage Account provides cloud-based storage for application files, documents and other data.


Create a storage account - Mic
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccountV4.ReactView
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage
Create a storage account
Encryption
Tags
Review + create
Project details
Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.
Subscription *
Azure subscription 1
Resource group *
rizzshaal
Create new
Instance details
Storage account name *
rizzwebstorage
Region *
(US) Central US
Deploy to an Azure Extended Zone


Create a storage account - Mic
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccountV4.ReactView
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage
Create a storage account
Primary service *
Azure Blob Storage or Azure Data Lake Storage
This selection helps us provide guidance as you create your storage account. It will not restrict you from using compatible storage services. Learn more
Primary workload
General purpose
Best for custom storage configurations, with non-workload-specific defaults
This selection applies settings to help you achieve optimal outcomes for your intended usage. You can further customize settings as you create your storage account. Compare workloads
Performance *
Standard
Recommended for most scenarios (general-purpose v2 account)
Premium
Recommended for scenarios that require low latency.
Standard performance is recommended for the "General purpose" workload.
Redundancy *
Locally redundant storage (LRS)


Create a storage account - Mic
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccountV4.ReactView
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage
Create a storage account
Basics
Advanced
Networking
Data protection
Security
Encryption
Tags
Review + create
Networking
Public access
Access your resource from anywhere through a public network.
Note: Allowing access to your resource through a public network increases security risk. Learn more
Public network access *
Enable
Allow inbound and outbound access with the option to restrict select inbound access using resource access configurations for this resource.
Disable
Restrict inbound access while allowing outbound access.
Secure by perimeter (Most restricted)
Restrict inbound and outbound access using a network security perimeter. Secure by perimeter offers the greatest level of inbound and outbound restriction to secure your resource.
Public network access scope *
Enable from all networks


Create a storage account - Mic
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccountV4.ReactView
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage
Create a storage account
Advanced
Networking
Data protection
Security
Encryption
Tags
Review + create
View automation template
Basics
Subscription : Azure subscription 1
Resource group : rizzshaal
Location : Central US
Storage account name : rizzwebstorage
Primary service : Azure Blob Storage or Azure Data Lake Storage
Primary workload : General purpose - Modified (Reset all to default)
These properties have been modified from the default recommendations:
• Redundancy: LRS (Recommended: RA-GRS)
Performance : Standard
Replication : Locally redundant storage (LRS)
Advanced
Enable hierarchical namespace : Disabled



Step 9: Create Blob Container and Upload File

Console path: Storage Account → Data Storage → Containers.

Implementation:

1. Open `rizzwebstorage`.
2. Select Containers.
3. Click + Container.
4. Enter container name `website-files`.

5. Set anonymous access to Private.
6. Click Create.

Create sample file on Windows:

Open Notepad and enter:

```
Welcome to Rizzshaal Cloud Project.

This file is stored in Azure Blob Storage.
Project: Website with Database and Cloud Storage.
Platform: Microsoft Azure.
```

Save as:

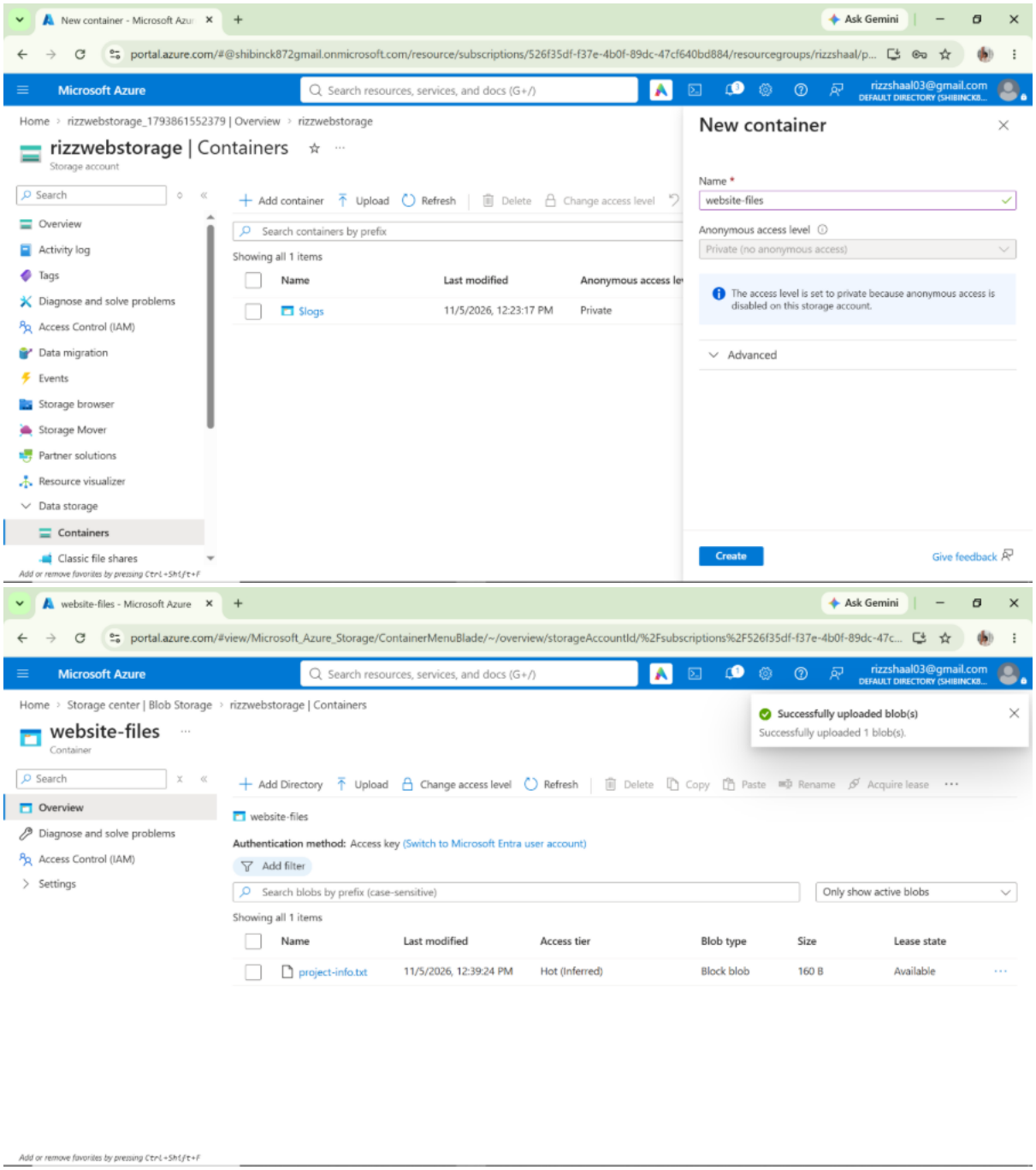
`project-info.txt`

Upload:

1. Open `website-files`.
2. Click Upload.
3. Select `project-info.txt`.
4. Click Upload.
5. Verify the file appears.

Explanation:

Blob Storage stores unstructured data such as text files, images, videos and documents.



Step 10: Configure Managed Identity

Console path: Azure Portal → Virtual Machines → `vm-rizz-web` → Identity.

Implementation:

1. Open System Assigned.
2. Enable Managed Identity.
3. Click Save.

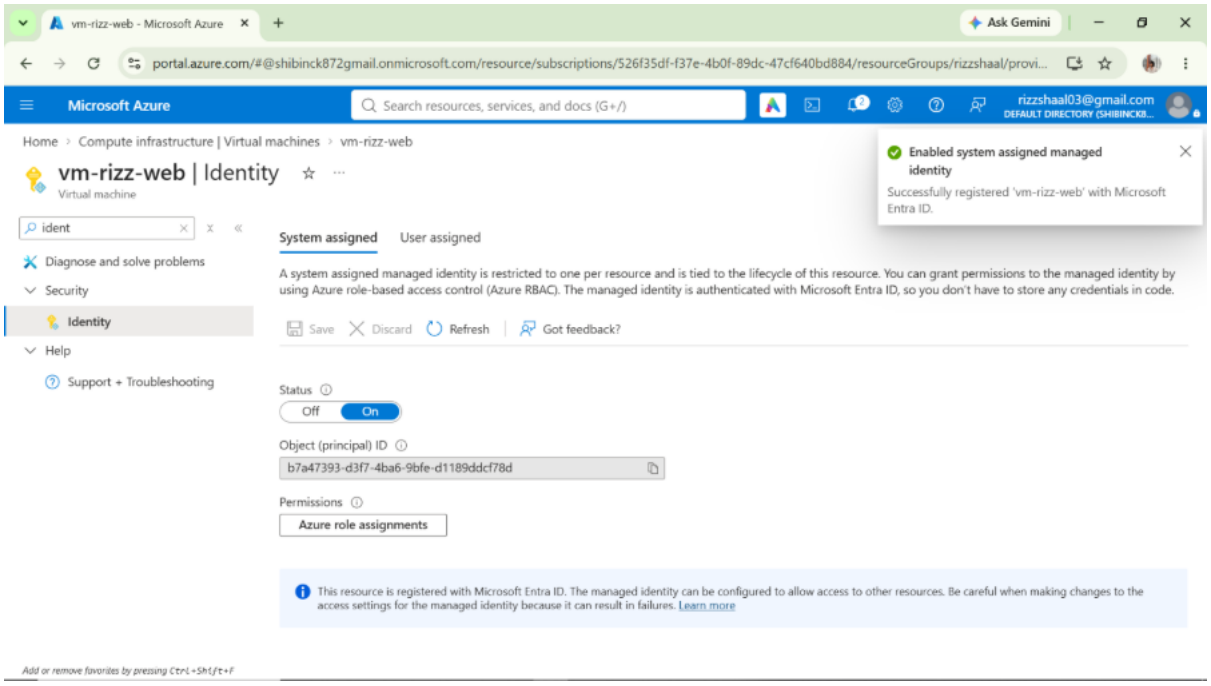
Assign storage permission:

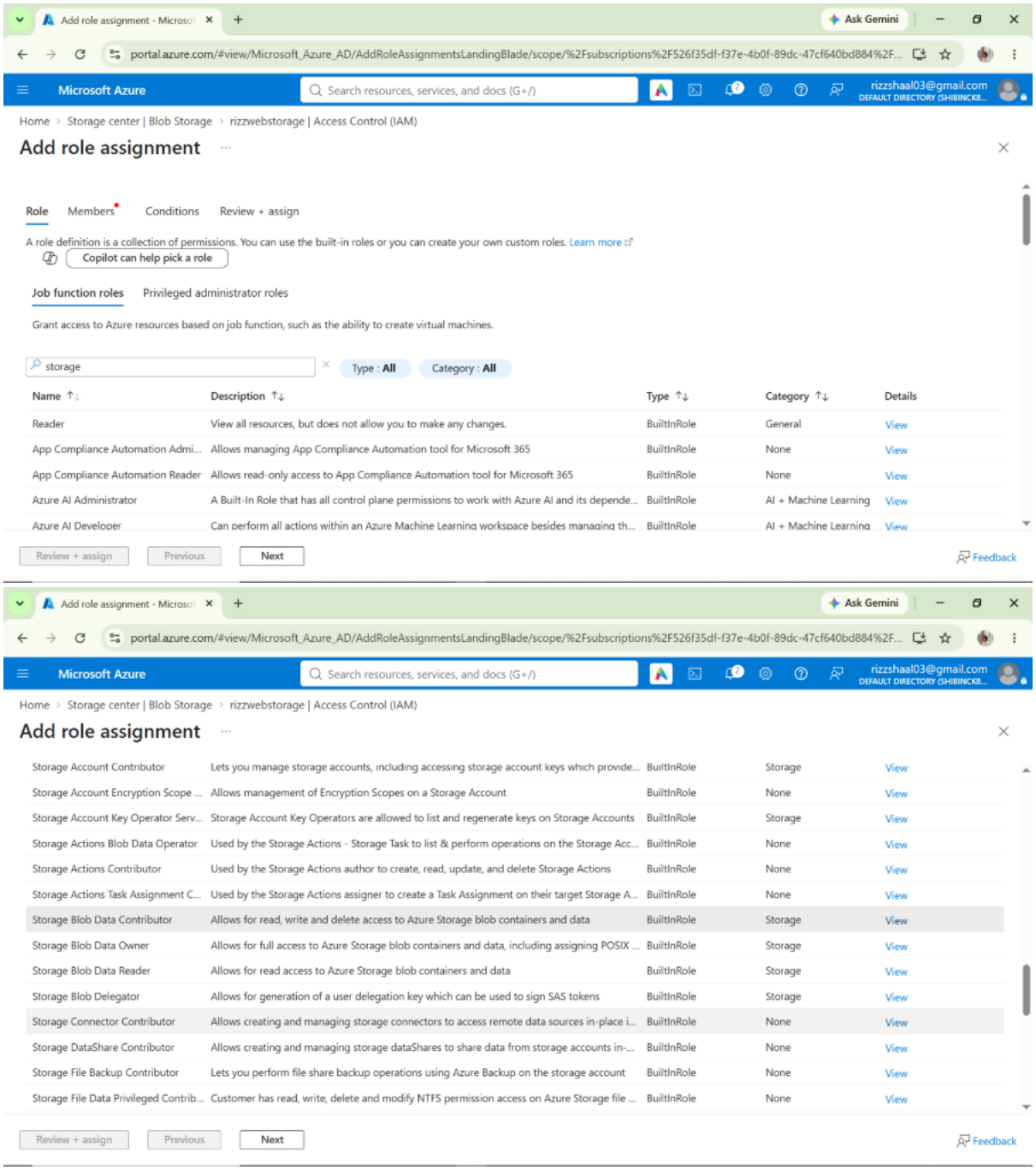
Console path: Storage Account → Access Control (IAM) → Add Role Assignment.

1. Select Add role assignment.
2. Search for Storage Blob Data Contributor.
3. Select the role.
4. Choose Managed Identity.
5. Select Virtual Machine.
6. Select `vm-rizz-web`.
7. Review and assign.

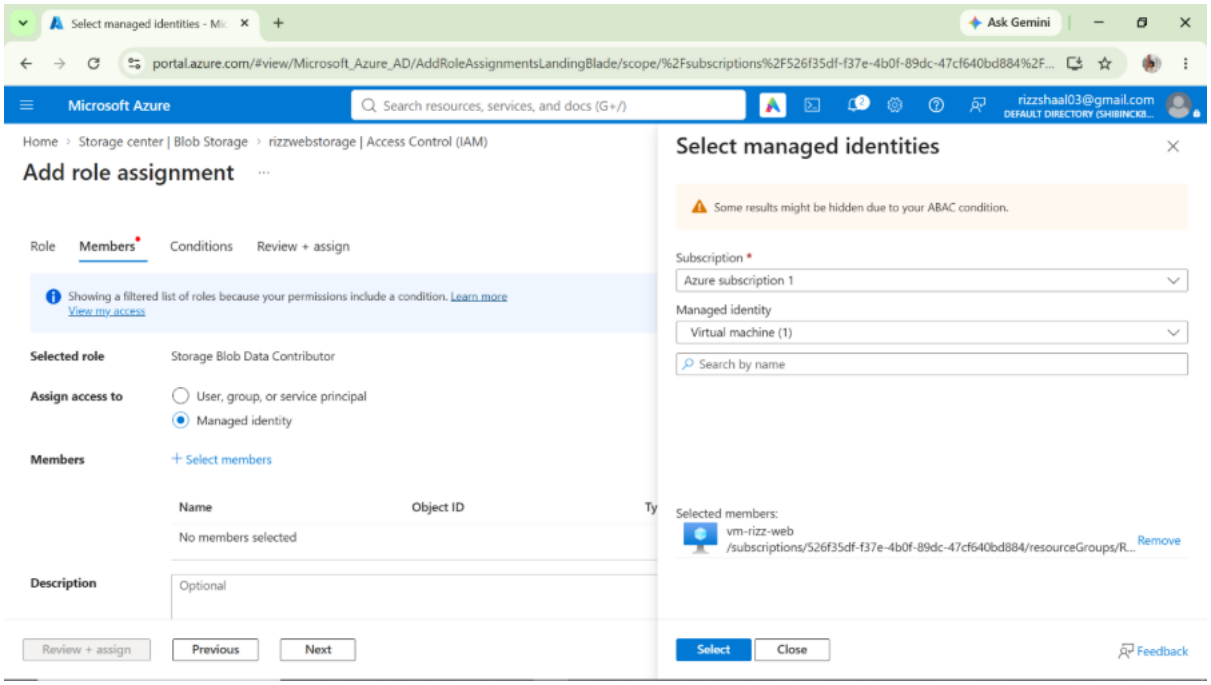
Explanation:

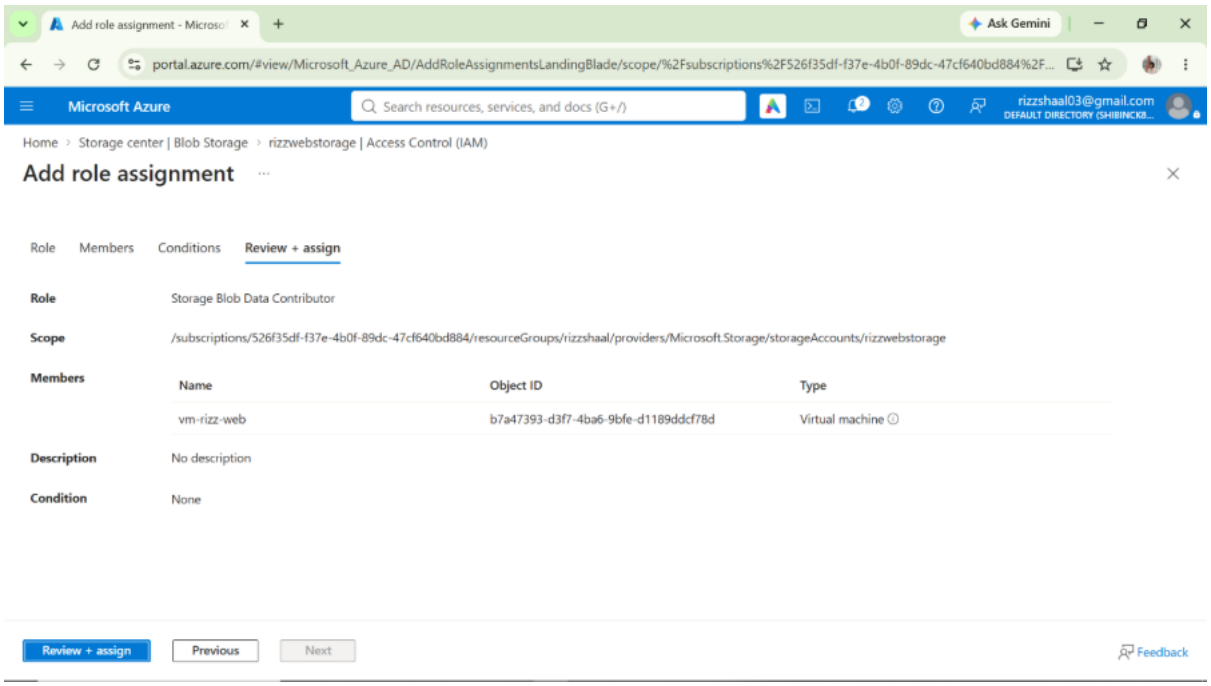
Managed Identity allows the VM to securely access Azure services without storing storage account credentials inside the application

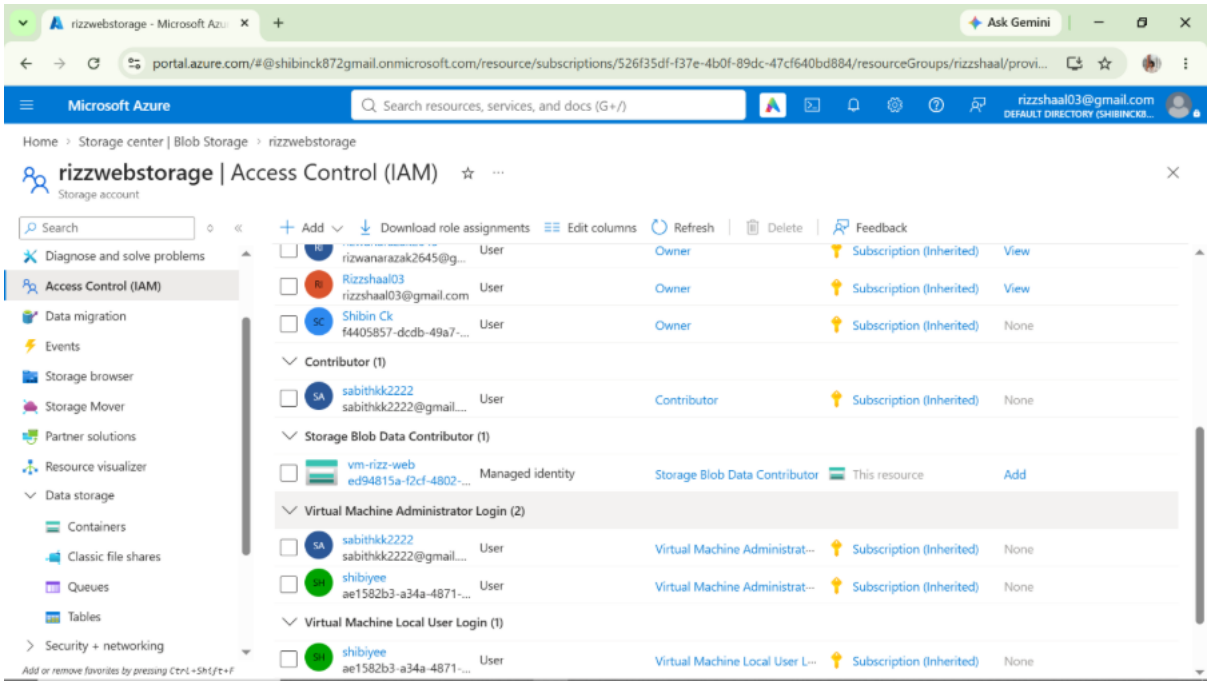



Add role assignment - Microso
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_AD/AddRoleAssignmentsLandingBlade/scope/%2Fsubscriptions%2F526f35df-f37e-4b0f-89dc-47cf640bd884%2F...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage › rizzwebstorage | Access Control (IAM)
Add role assignment
Role
Members
Conditions
Review + assign
A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. Learn more
Copilot can help pick a role
Job function roles
Privileged administrator roles
Grant access to Azure resources based on job function, such as the ability to create virtual machines.
storage
Type : All
Category : All
Name
Description
Type
Category
Details
Reader
View all resources, but does not allow you to make any changes.
BuiltInRole
General
View
App Compliance Automation Admi...
Allows managing App Compliance Automation tool for Microsoft 365
BuiltInRole
None
View
App Compliance Automation Reader
Allows read-only access to App Compliance Automation tool for Microsoft 365
BuiltInRole
None
View
Azure AI Administrator
A Built-In Role that has all control plane permissions to work with Azure AI and its depende...
BuiltInRole
AI + Machine Learning
View
Azure AI Developer
Can perform all actions within an Azure Machine Learning workspace besides managing th...
BuiltInRole
AI + Machine Learning
View
Review + assign
Previous
Next
Feedback

Add role assignment - Microso
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_AD/AddRoleAssignmentsLandingBlade/scope/%2Fsubscriptions%2F526f35df-f37e-4b0f-89dc-47cf640bd884%2F...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage › rizzwebstorage | Access Control (IAM)
Add role assignment
Storage Account Contributor
Lets you manage storage accounts, including accessing storage account keys which provide...
BuiltInRole
Storage
View
Storage Account Encryption Scope ...
Allows management of Encryption Scopes on a Storage Account
BuiltInRole
None
View
Storage Account Key Operator Serv...
Storage Account Key Operators are allowed to list and regenerate keys on Storage Accounts
BuiltInRole
Storage
View
Storage Actions Blob Data Operator
Used by the Storage Actions - Storage Task to list & perform operations on the Storage Acc...
BuiltInRole
None
View
Storage Actions Contributor
Used by the Storage Actions author to create, read, update, and delete Storage Actions
BuiltInRole
None
View
Storage Actions Task Assignment C...
Used by the Storage Actions assigner to create a Task Assignment on their target Storage A...
BuiltInRole
None
View
Storage Blob Data Contributor
Allows for read, write and delete access to Azure Storage blob containers and data
BuiltInRole
Storage
View
Storage Blob Data Owner
Allows for full access to Azure Storage blob containers and data, including assigning POSIX ...
BuiltInRole
Storage
View
Storage Blob Data Reader
Allows for read access to Azure Storage blob containers and data
BuiltInRole
Storage
View
Storage Blob Delegator
Allows for generation of a user delegation key which can be used to sign SAS tokens
BuiltInRole
Storage
View
Storage Connector Contributor
Allows creating and managing storage connectors to access remote data sources in-place i...
BuiltInRole
None
View
Storage DataShare Contributor
Allows creating and managing storage dataShares to share data from storage accounts in-...
BuiltInRole
None
View
Storage File Backup Contributor
Lets you perform file share backup operations using Azure Backup on the storage account
BuiltInRole
None
View
Storage File Data Privileged Contrib...
Customer has read, write, delete and modify NTFS permission access on Azure Storage file ...
BuiltInRole
None
View
Review + assign
Previous
Next
Feedback


Select managed identities - Mic
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_AD/AddRoleAssignmentsLandingBlade/scope/%2Fsubscriptions%2F526f35df-f37e-4b0f-89dc-47cf640bd884%2F...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage › rizzwebstorage | Access Control (IAM)
Add role assignment
Role
Members
Conditions
Review + assign
Showing a filtered list of roles because your permissions include a condition. Learn more
View my access
Selected role
Storage Blob Data Contributor
Assign access to
User, group, or service principal
Managed identity
Members
+ Select members
Name
Object ID
Ty
No members selected
Description
Optional
Review + assign
Previous
Next
Select managed identities
Some results might be hidden due to your ABAC condition.
Subscription *
Azure subscription 1
Managed identity
Virtual machine (1)
Search by name
Selected members:
vm-rizz-web
/subscriptions/526f35df-f37e-4b0f-89dc-47cf640bd884/resourceGroups/R...
Remove
Select
Close
Feedback


Add role assignment - Microso
Ask Gemini
portal.azure.com/#view/Microsoft_Azure_AD/AddRoleAssignmentsLandingBlade/scope/%2Fsubscriptions%2F526f35df-f37e-4b0f-89dc-47cf640bd884%2F...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Storage center | Blob Storage › rizzwebstorage | Access Control (IAM)
Add role assignment
Role
Members
Conditions
Review + assign
Role
Storage Blob Data Contributor
Scope
/subscriptions/526f35df-f37e-4b0f-89dc-47cf640bd884/resourceGroups/rizzshaal/providers/Microsoft.Storage/storageAccounts/rizzwebstorage
Members
Name
Object ID
Type
vm-rizz-web
b7a47393-d3f7-4ba6-9bfe-d1189ddcf78d
Virtual machine
Description
No description
Condition
None
Review + assign
Previous
Next
Feedback



Step 11: Connect Linux VM to Blob Storage

Console path: Ubuntu SSH Terminal.

Check Azure CLI:

```
az --version
```

Install if required:

```
curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash
```

Authenticate:

```
az login --identity
```

Check account:

```
az account show
```

Download the uploaded file:

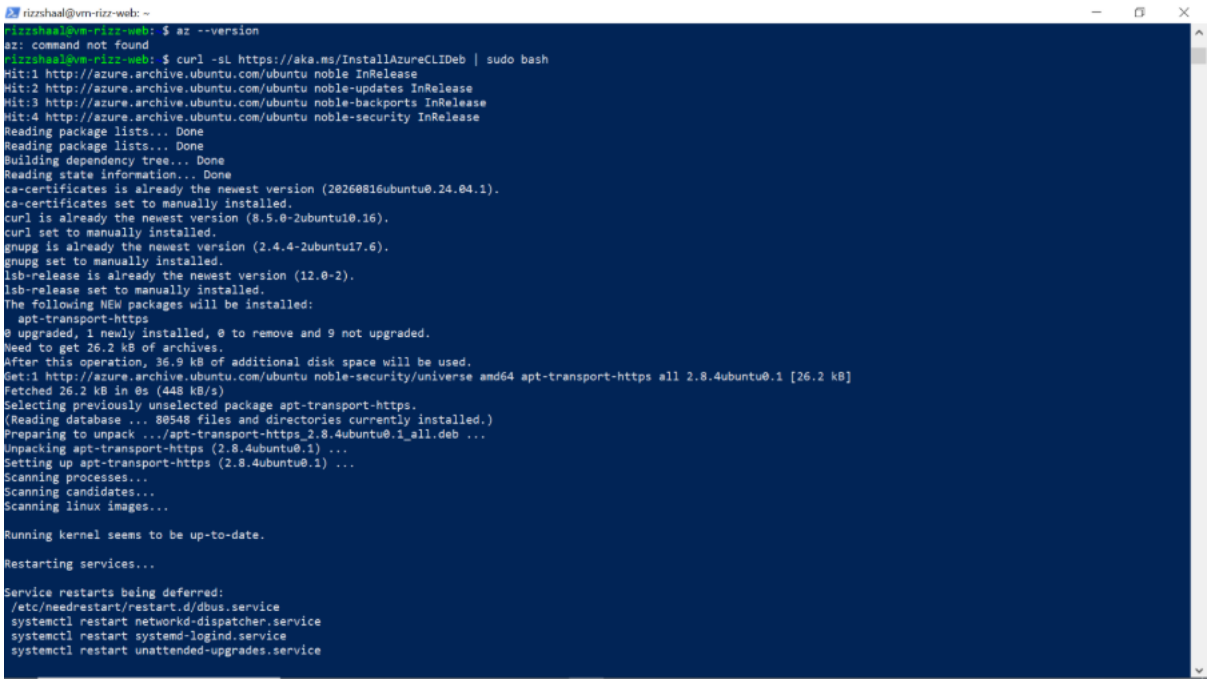
```
az storage blob download \
--account-name rizzwebstorage \
--container-name website-files \
--name project-info.txt \
--file ~/project-info.txt \
--auth-mode login
```

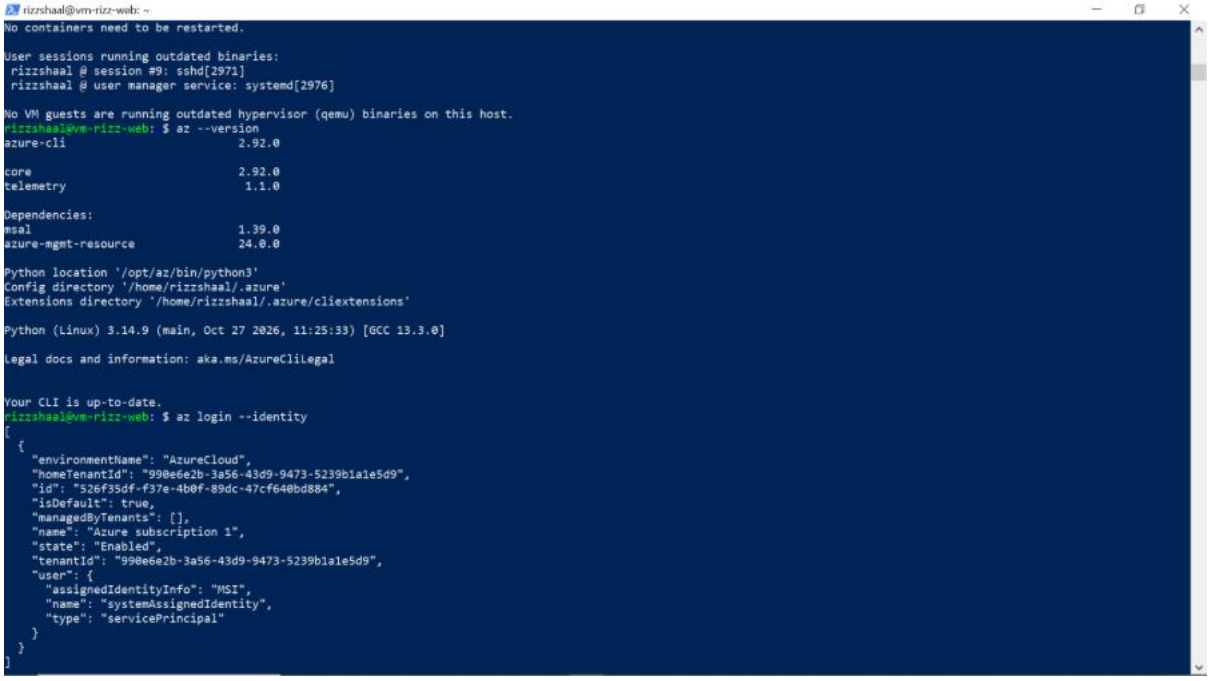
Verify:

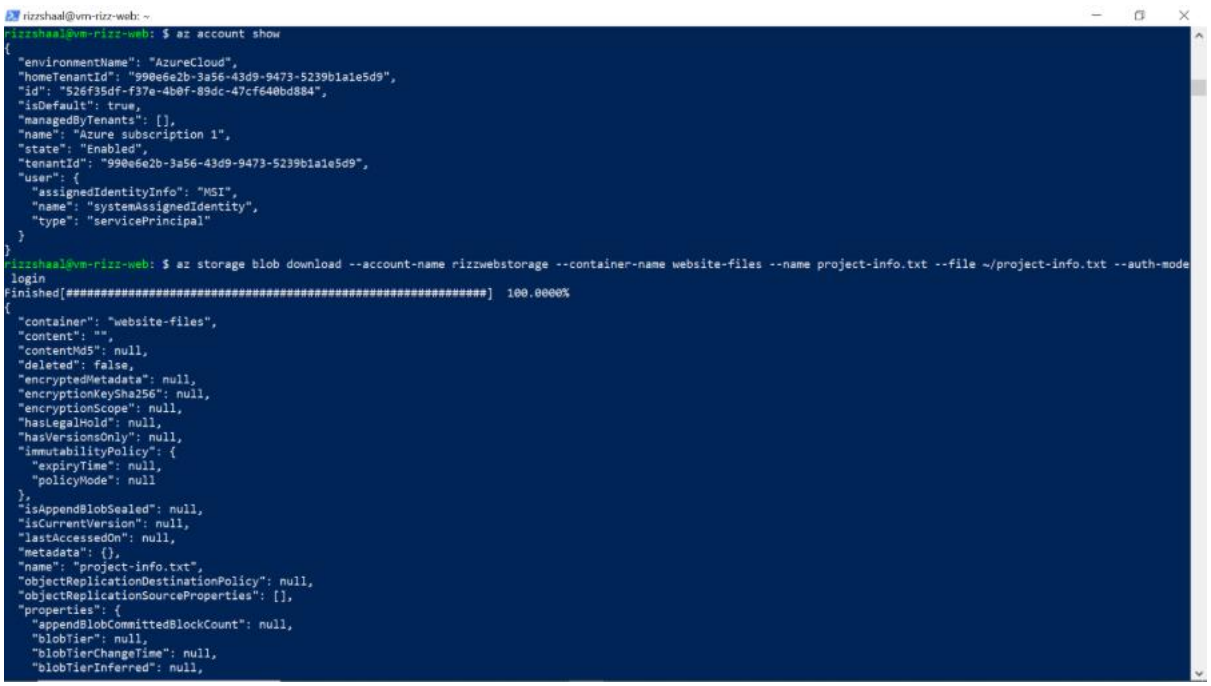
```
cat ~/project-info.txt
```

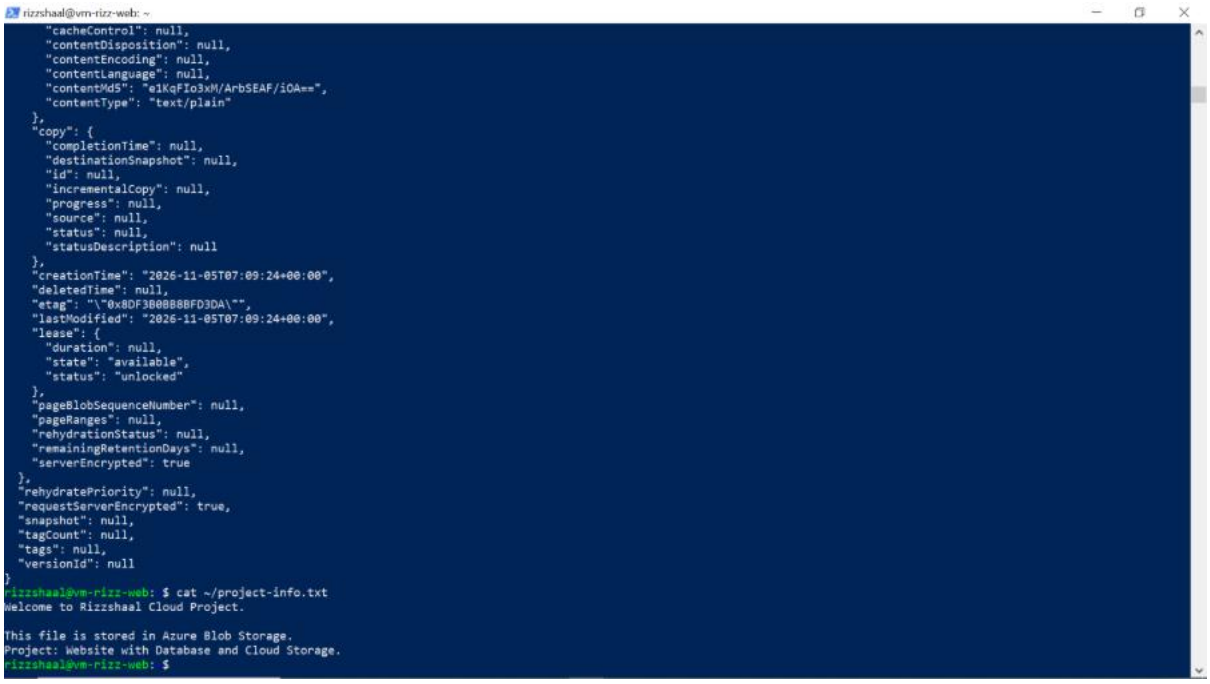
Explanation:

This step connects the Linux VM to Azure Blob Storage using Managed Identity and verifies file retrieval.




rizzshaal@vm-rizz-web: ~
No containers need to be restarted.
User sessions running outdated binaries:
rizzshaal @ session #9: sshd[2971]
rizzshaal @ user manager service: systemd[2976]
No VM guests are running outdated hypervisor (qemu) binaries on this host.
rizzshaal@vm-rizz-web: $ az --version
azure-cli 2.92.0
core 2.92.0
telemetry 1.1.0
Dependencies:
msal 1.39.0
azure-mgmt-resource 24.0.0
Python location '/opt/az/bin/python3'
Config directory '/home/rizzshaal/.azure'
Extensions directory '/home/rizzshaal/.azure/cliextensions'
Python (Linux) 3.14.9 (main, Oct 27 2026, 11:25:33) [GCC 13.3.0]
Legal docs and information: aka.ms/AzureCliLegal
Your CLI is up-to-date.
rizzshaal@vm-rizz-web: $ az login --identity
[
{
"environmentName": "AzureCloud",
"homeTenantId": "990e6e2b-3a56-43d9-9473-5239b1a1e5d9",
"id": "526f35df-f37e-4b0f-89dc-47cf640bd884",
"isDefault": true,
"managedByTenants": [],
"name": "Azure subscription 1",
"state": "Enabled",
"tenantId": "990e6e2b-3a56-43d9-9473-5239b1a1e5d9",
"user": {
"assignedIdentityInfo": "MSI",
"name": "systemAssignedIdentity",
"type": "servicePrincipal"
}
}
]


rizzshaal@vm-rizz-web: ~
rizzshaal@vm-rizz-web: $ az account show
{
"environmentName": "AzureCloud",
"homeTenantId": "990e6e2b-3a56-43d9-9473-5239b1a1e5d9",
"id": "526f35df-f37e-4b0f-89dc-47cf640bd884",
"isDefault": true,
"managedByTenants": [],
"name": "Azure subscription 1",
"state": "Enabled",
"tenantId": "990e6e2b-3a56-43d9-9473-5239b1a1e5d9",
"user": {
"assignedIdentityInfo": "MSI",
"name": "systemAssignedIdentity",
"type": "servicePrincipal"
}
}
rizzshaal@vm-rizz-web: $ az storage blob download --account-name rizzwebstorage --container-name website-files --name project-info.txt --file ~/project-info.txt --auth-mode login
Finished[###] 100.0000%
{
"container": "website-files",
"content": "",
"contentMd5": null,
"deleted": false,
"encryptedMetadata": null,
"encryptionKeySha256": null,
"encryptionScope": null,
"hasLegalHold": null,
"hasVersionsOnly": null,
"immutabilityPolicy": {
"expiryTime": null,
"policyMode": null
},
"isAppendBlobSealed": null,
"isCurrentVersion": null,
"lastAccessedOn": null,
"metadata": {},
"name": "project-info.txt",
"objectReplicationDestinationPolicy": null,
"objectReplicationSourceProperties": [],
"properties": {
"appendBlobCommittedBlockCount": null,
"blobTier": null,
"blobTierChangeTime": null,
"blobTierInferred": null,



Step 12: Prepare Website Application

Console path: Ubuntu SSH Terminal → Website project directory.

The website application contains the following files:

File	Purpose
index.html	Website homepage
style.css	Website styling
script.js	Website functionality
db.php	Database connection
users.php	Displays database records

Install PHP and the MySQL extension if required:

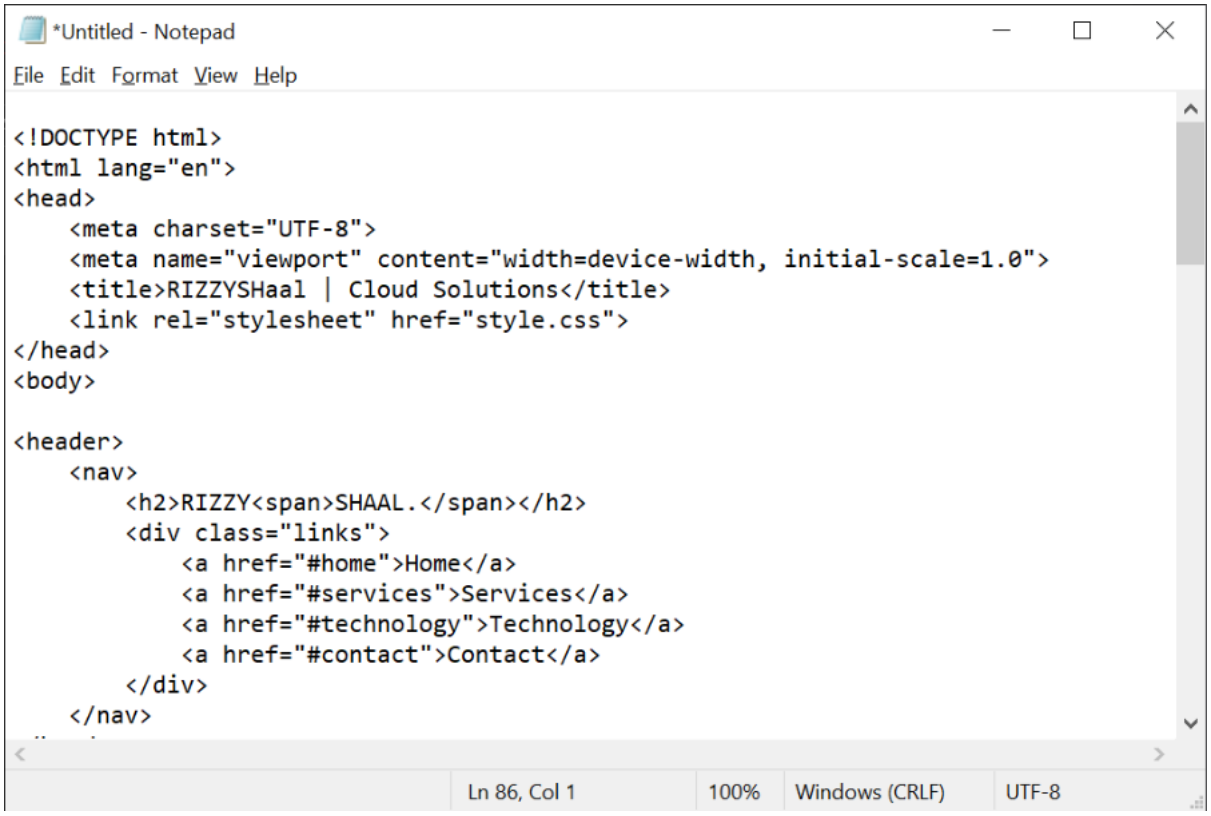
```
sudo apt install php libapache2-mod-php php-mysql -y
```

Create the website directory:

```
sudo mkdir -p /var/www/rizz-cloud-website
```

Explanation:

PHP allows the website to communicate with MySQL and retrieve database information.



db - Notepad

File Edit Format View Help

```php
<?php
$host = "YOUR_MYSQL_HOSTNAME";
$user = "mysqladmin";
$password = "YOUR_MYSQL_PASSWORD";
$database = "website_db";

$conn = new mysqli($host, $user, $password, $database);

if ($conn->connect_error) {
    die("Database connection failed");
}

$conn->set_charset("utf8mb4");
?>
```

Ln 16, Col 1 | 100% | Windows (CRLF) | UTF-8

users - Notepad

File Edit Format View Help

```php
    echo json_encode(["message" => "Enter a valid name and email"]);
    exit;
}

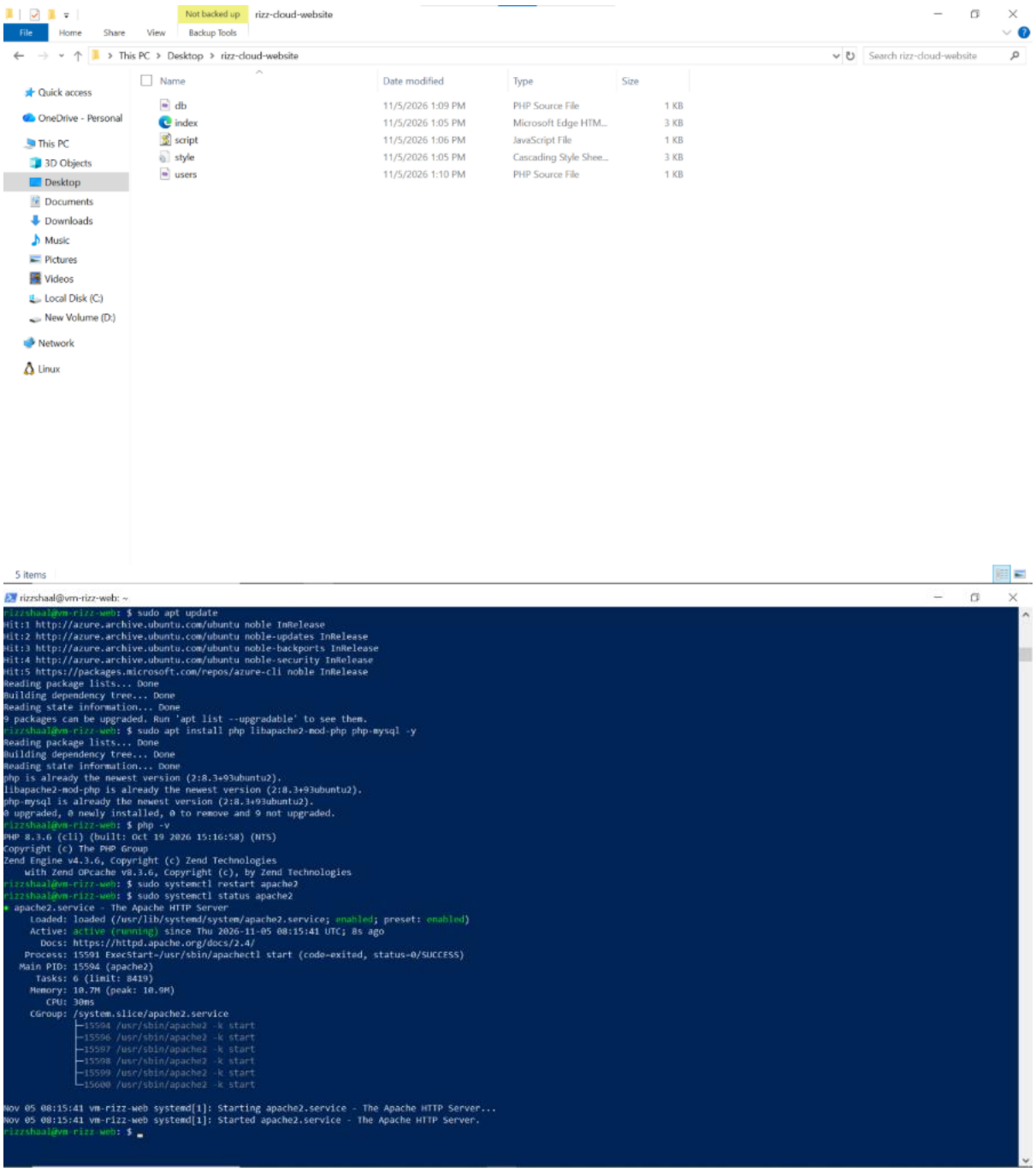
$stmt = $conn->prepare(
    "INSERT INTO users (name, email) VALUES (?, ?)"
);

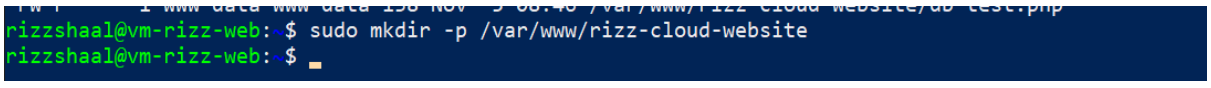
$stmt->bind_param("ss", $name, $email);

if ($stmt->execute()) {
    echo json_encode(["message" => "Details saved successfully!"]);
} else {
    http_response_code(500);
    echo json_encode(["message" => "Unable to save details"]);
}

$stmt->close();
$conn->close();
?>
```

Ln 40, Col 1 | 100% | Windows (CRLF) | UTF-8





Step 13: Configure Database Connection

Console path: Ubuntu SSH Terminal → `/etc/`.

Create a protected database configuration file:

```
sudo nano /etc/rizz-cloud-db.conf
```

Add the following configuration using your actual database credentials:

```
DB_HOST=mysql-rizz-web.mysql.database.azure.com
DB_USER=mysqladmin
DB_PASS=YOUR_MYSQL_PASSWORD
DB_NAME=website_db
```

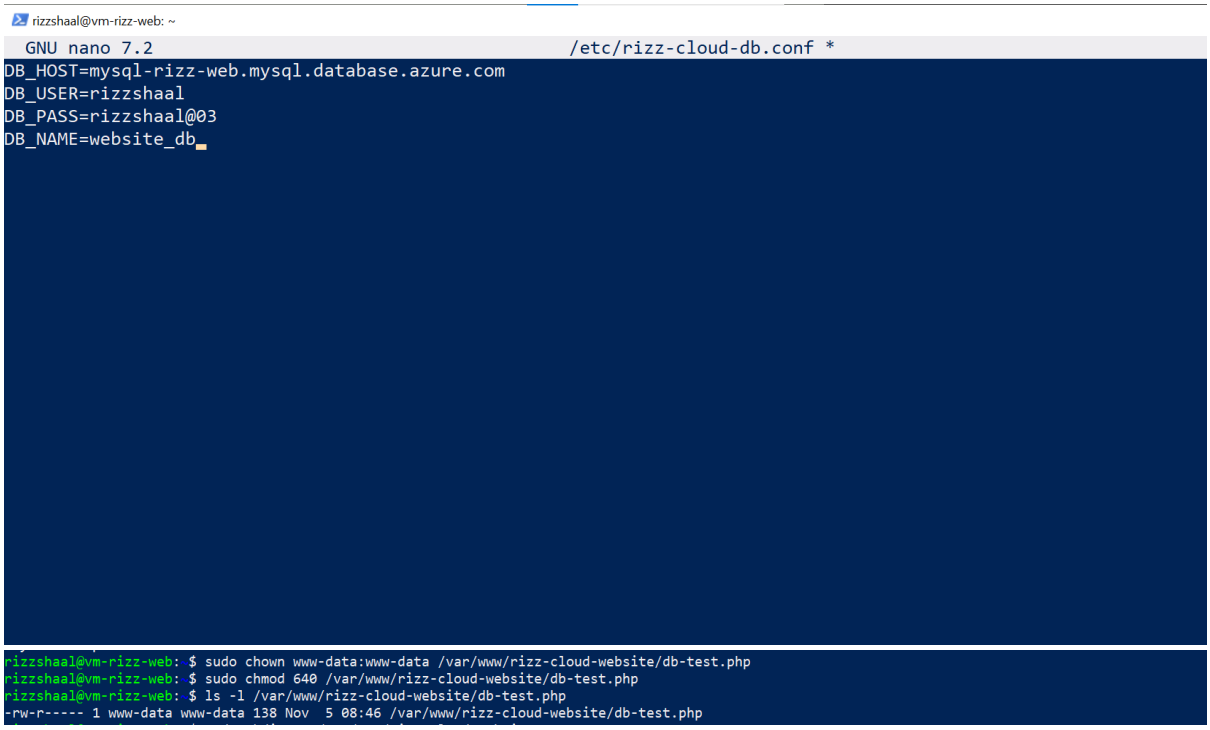
Save the file.

Set secure ownership and permissions:

```
sudo chown root:www-data /etc/rizz-cloud-db.conf
sudo chmod 640 /etc/rizz-cloud-db.conf
```

Explanation:

This configuration stores database connection details separately from the publicly accessible website files.



Step 14: Configure Website Database Integration

Console path: Ubuntu SSH Terminal → `/var/www/rizz-cloud-website/`.

Create or configure `db.php` to read the protected database configuration.

Example:

```
<?php
$config = parse_ini_file('/etc/rizz-cloud-db.conf');

$conn = new mysqli(
    $config['DB_HOST'],
    $config['DB_USER'],
    $config['DB_PASS'],
    $config['DB_NAME']
);

if ($conn->connect_error) {
    die("Database connection failed");
}
?>
```

Create `users.php` to retrieve database records:

```
<?php
require 'db.php';

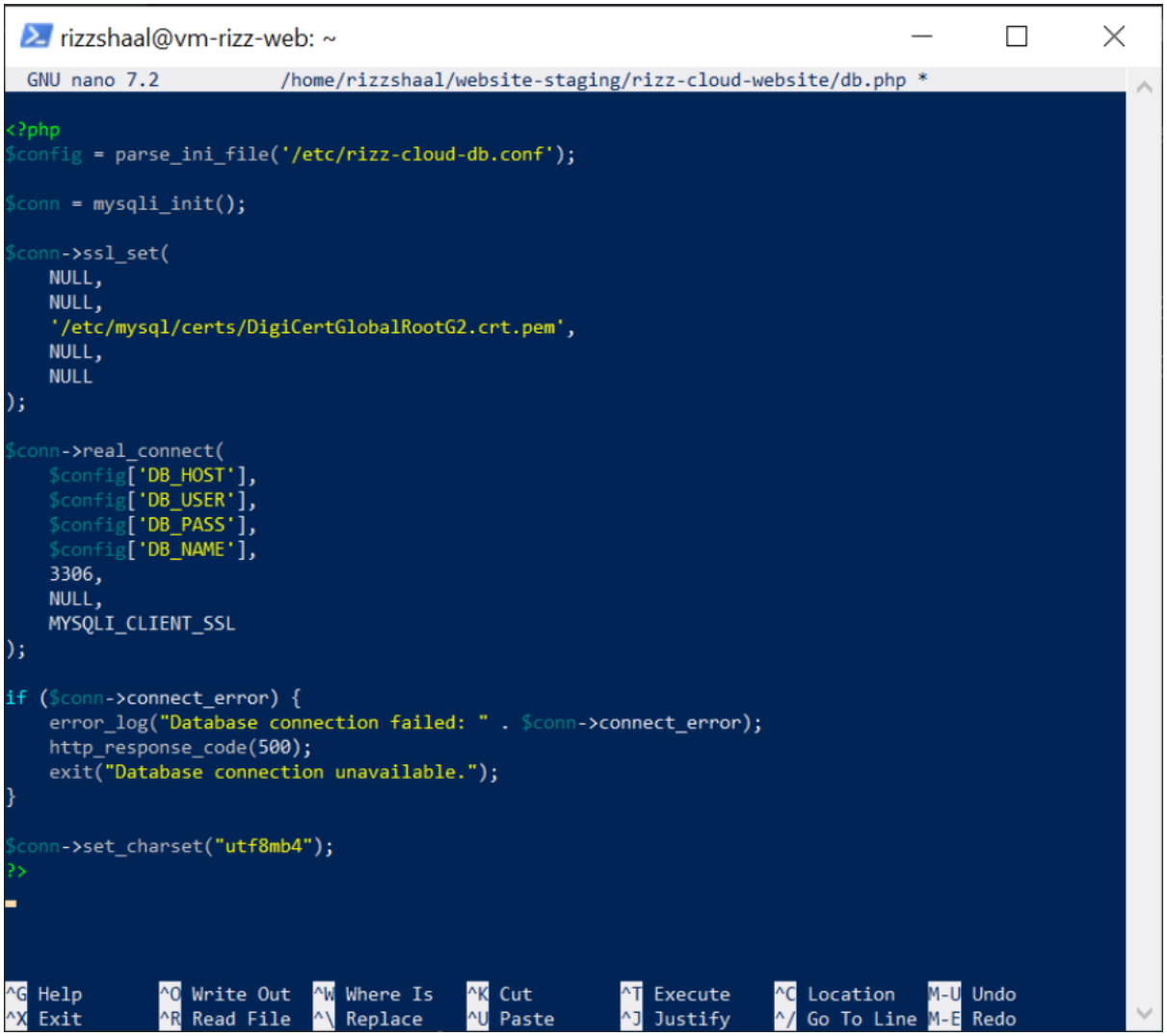
$result = $conn->query("SELECT * FROM users");

while ($row = $result->fetch_assoc()) {
    echo htmlspecialchars($row['name']) . " - ";
    echo htmlspecialchars($row['email']) . "<br>";
}

$conn->close();
?>
```

Explanation:

The PHP application connects to Azure MySQL and retrieves records from the users table.



Step 15: Compress Website Source Code into ZIP

Mandatory assignment requirement

Console path: Windows → Website project folder.

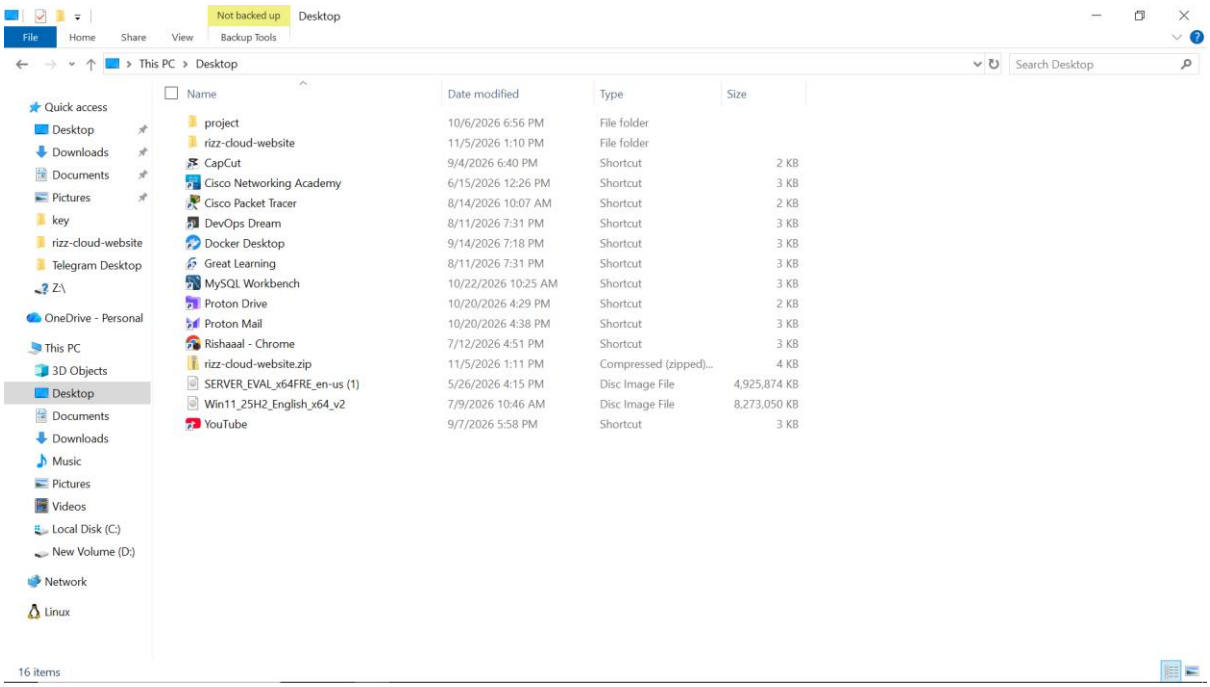
Implementation:

1. Locate the website source files on your computer.
2. Select the application files.
3. Right-click.
4. Select Send to → Compressed (zipped) folder.
5. Save the ZIP file as:

`rizz-cloud-website.zip`

Explanation:

Compressing the source code into ZIP format makes it easier to transfer the application from the host computer to the Azure VM.



Step 16: Transfer ZIP File to Linux VM

Console path: Windows PowerShell.

Use SCP:

```
scp -i "C:\path\to\rizz-web-key.pem"
"$env:USERPROFILE\Downloads\rizz-cloud-website.zip"
azureuser@YOUR_PUBLIC_IP:~/
```

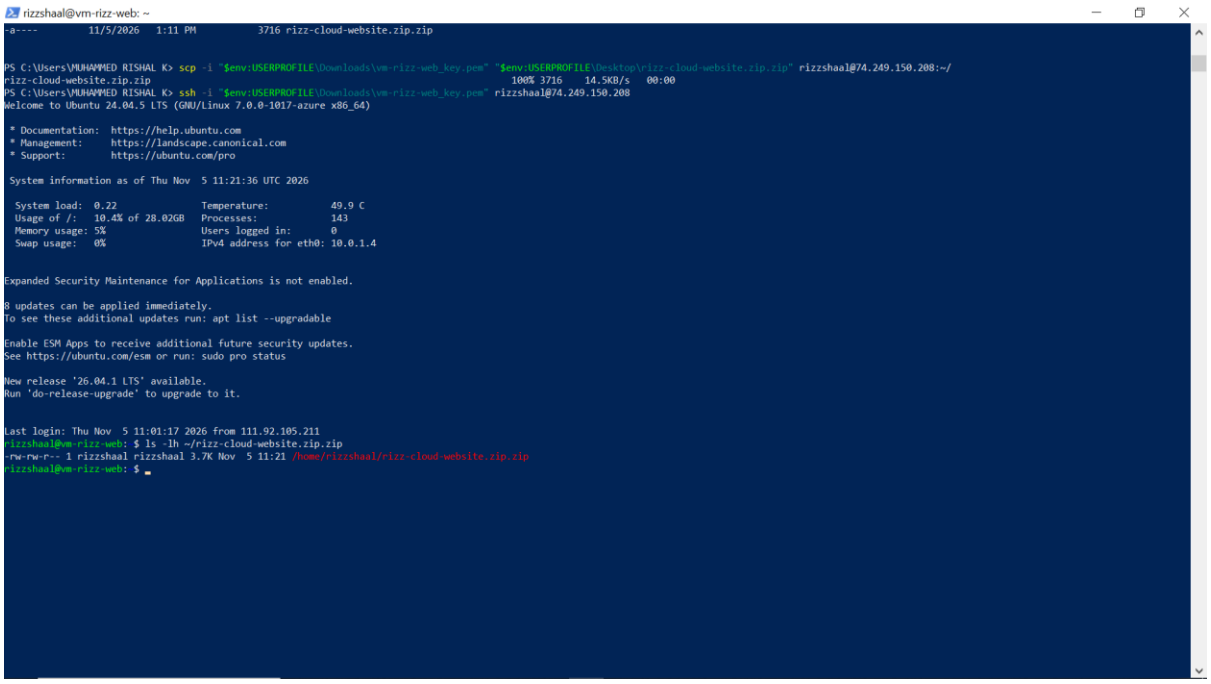
Replace the ZIP path and public IP with the actual values.

Verify on Ubuntu:

```
ls -lh ~/rizz-cloud-website.zip
```

Explanation:

SCP securely transfers the compressed website application from the host computer to the Azure Linux VM.



Step 17: Extract and Deploy Website

Console path: Ubuntu SSH Terminal.

Install unzip if necessary:

```
sudo apt install unzip -y
```

Create a temporary extraction directory:

```
mkdir -p ~/website-extract
```

Extract the ZIP file:

```
unzip ~/rizz-cloud-website.zip -d ~/website-extract
```

Inspect the extracted files:

```
find ~/website-extract -maxdepth 2 -type f
```

Copy the application files into the Apache website directory after verifying the extracted folder structure:

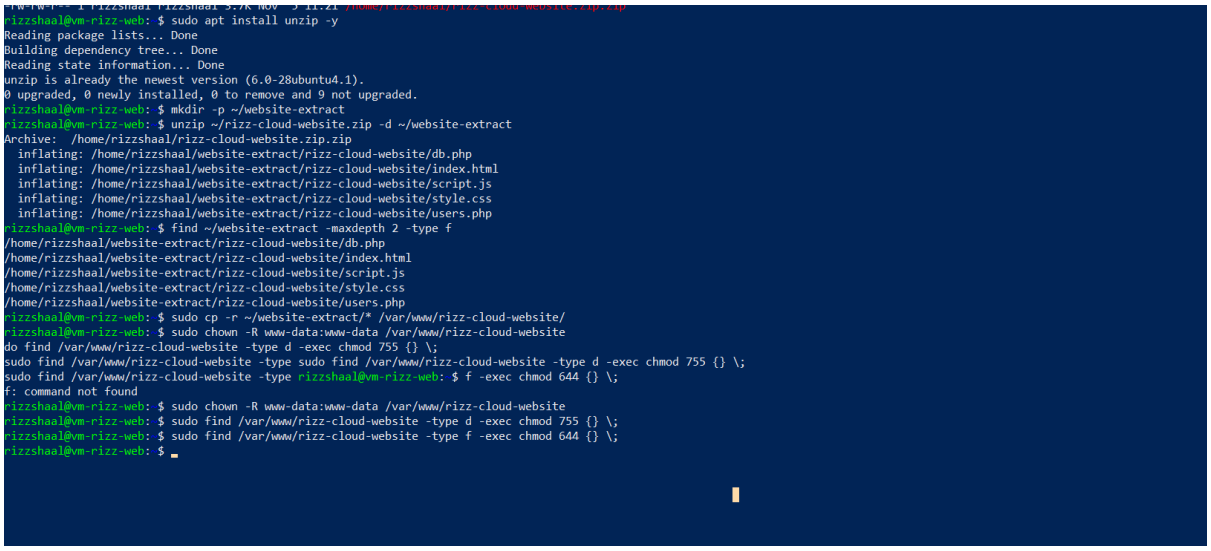
```
sudo cp -r ~/website-extract/* /var/www/rizz-cloud-website/
```

Configure permissions:

```
sudo chown -R www-data:www-data /var/www/rizz-cloud-website
sudo find /var/www/rizz-cloud-website -type d -exec chmod 755 {} \;
sudo find /var/www/rizz-cloud-website -type f -exec chmod 644 {} \;
```

Explanation:

The ZIP file is extracted first, and the application files are then deployed to the Apache website directory. This satisfies the assignment's source-code transfer requirement.



Step 18: Configure Apache Website

Console path: Ubuntu SSH Terminal → Apache configuration.

Create a virtual host:

```
sudo nano /etc/apache2/sites-available/rizz-cloud-website.conf
```

Add:

```
<VirtualHost *:80>
    ServerName localhost
    DocumentRoot /var/www/rizz-cloud-website

    <Directory /var/www/rizz-cloud-website>
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/rizz-error.log
```

```
    CustomLog ${APACHE_LOG_DIR}/rizz-access.log combined
</VirtualHost>
```

Enable the website:

```
sudo a2ensite rizz-cloud-website.conf
```

Disable the default website if it conflicts:

```
sudo a2dissite 000-default.conf
```

Test Apache configuration:

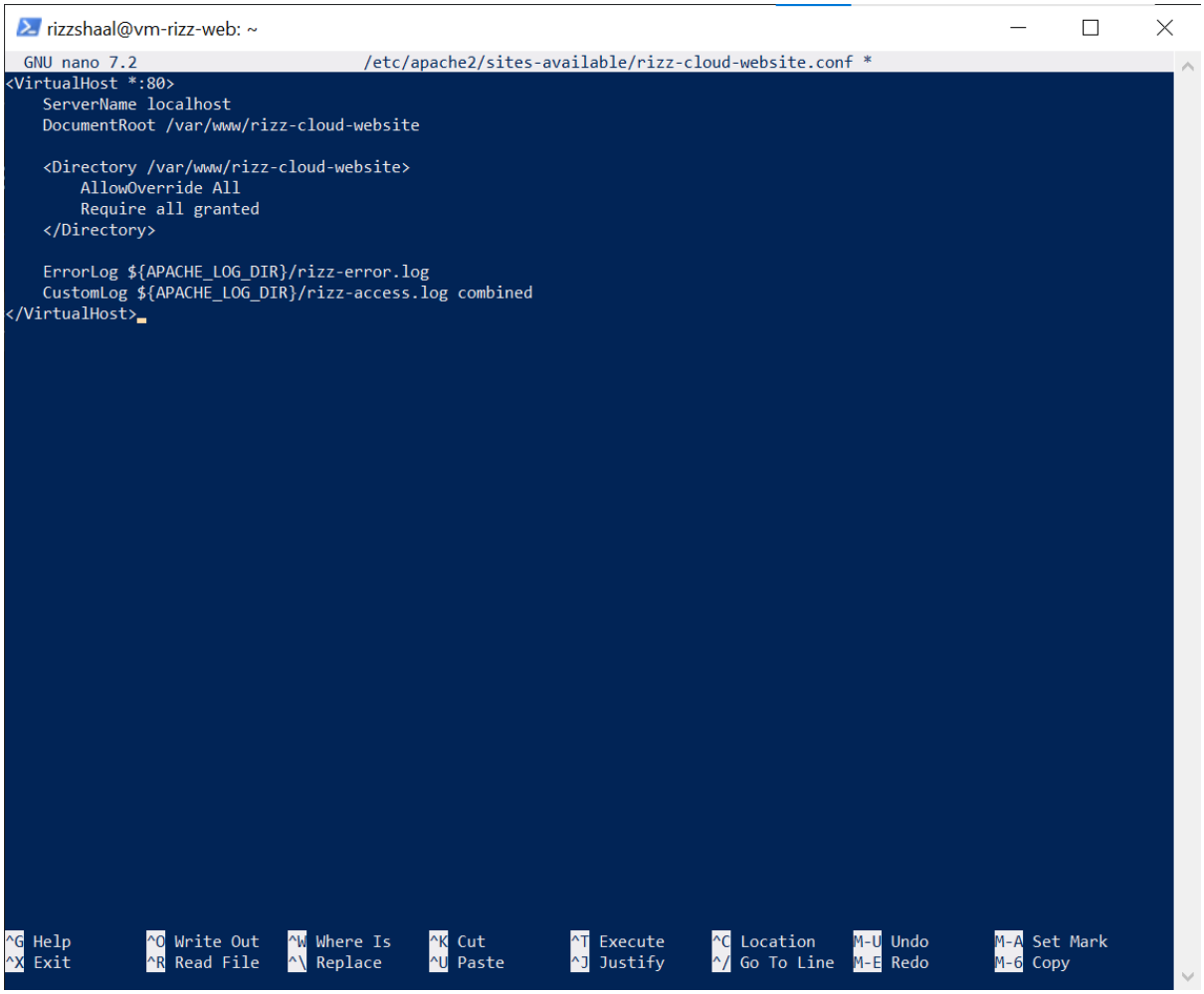
```
sudo apache2ctl configtest
```

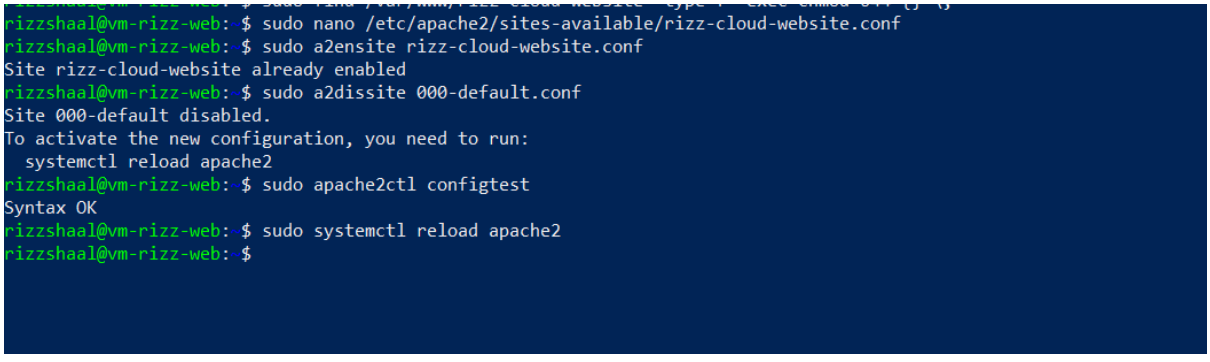
Restart Apache:

```
sudo systemctl reload apache2
```

Explanation:

Apache is configured to serve the deployed application from the custom website directory.





Step 19: Configure Storage Integration with Website

Console path: Ubuntu SSH Terminal → Azure CLI.

Use the VM's Managed Identity to retrieve files from Blob Storage.

For example:

```
az storage blob download \
--account-name rizzwebstorage \
```

```
--container-name website-files \
--name project-info.txt \
--file /tmp/project-info.txt \
--auth-mode login
```

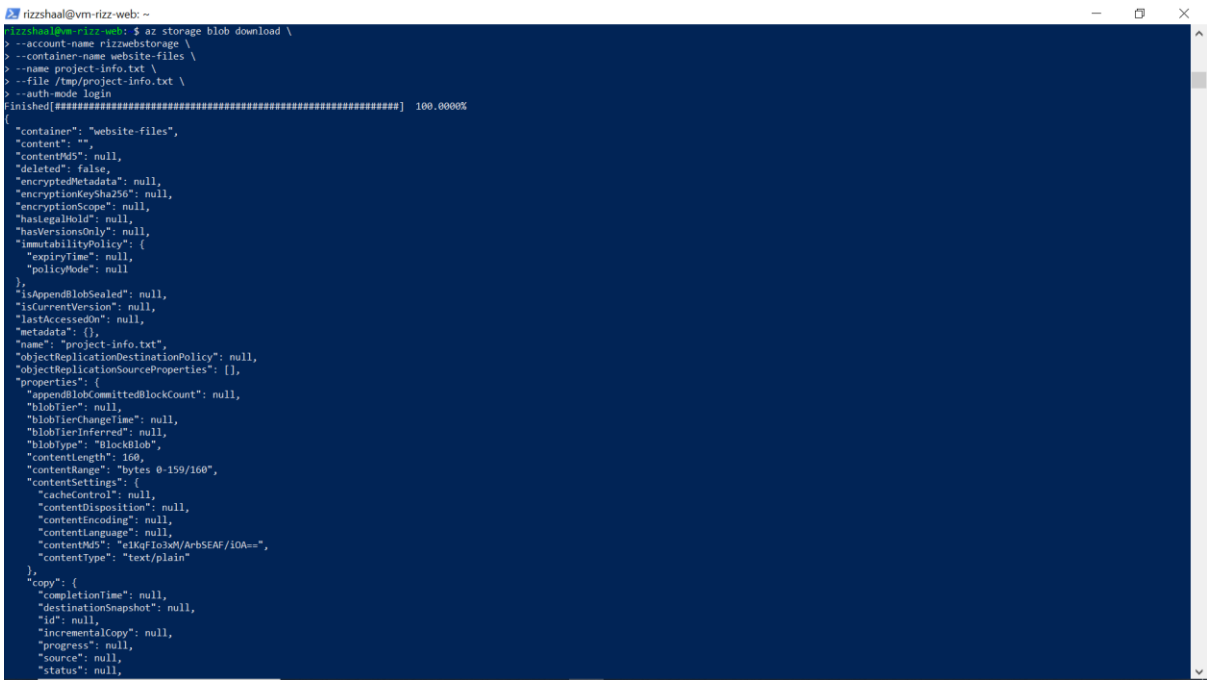
Verify:

```
cat /tmp/project-info.txt
```

For application integration, the PHP application can use Azure Blob Storage SDK or an authenticated backend process to retrieve files.

Explanation:

This demonstrates how the application environment can access files stored in Azure Blob Storage securely.





Step 20: Test Website Accessibility

Console path: Web browser → Address bar.

Enter:

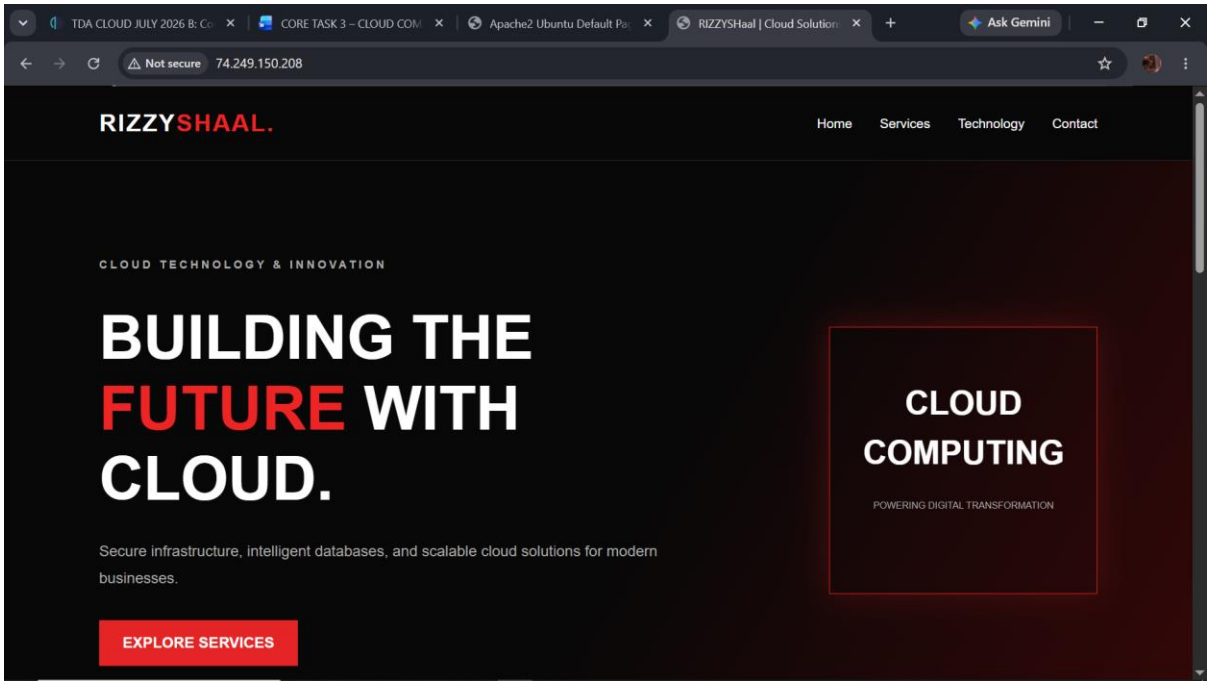
http://YOUR_PUBLIC_IP

Verify:

- Website homepage loads.
- CSS styling works.
- JavaScript functionality works.
- PHP pages execute correctly.

Explanation:

This test verifies that Apache successfully hosts the deployed website and that users can access it through the VM's public IP address.



Step 21: Test Database Connectivity

Console path: Browser → Website.

Open:

http://YOUR_PUBLIC_IP/users.php

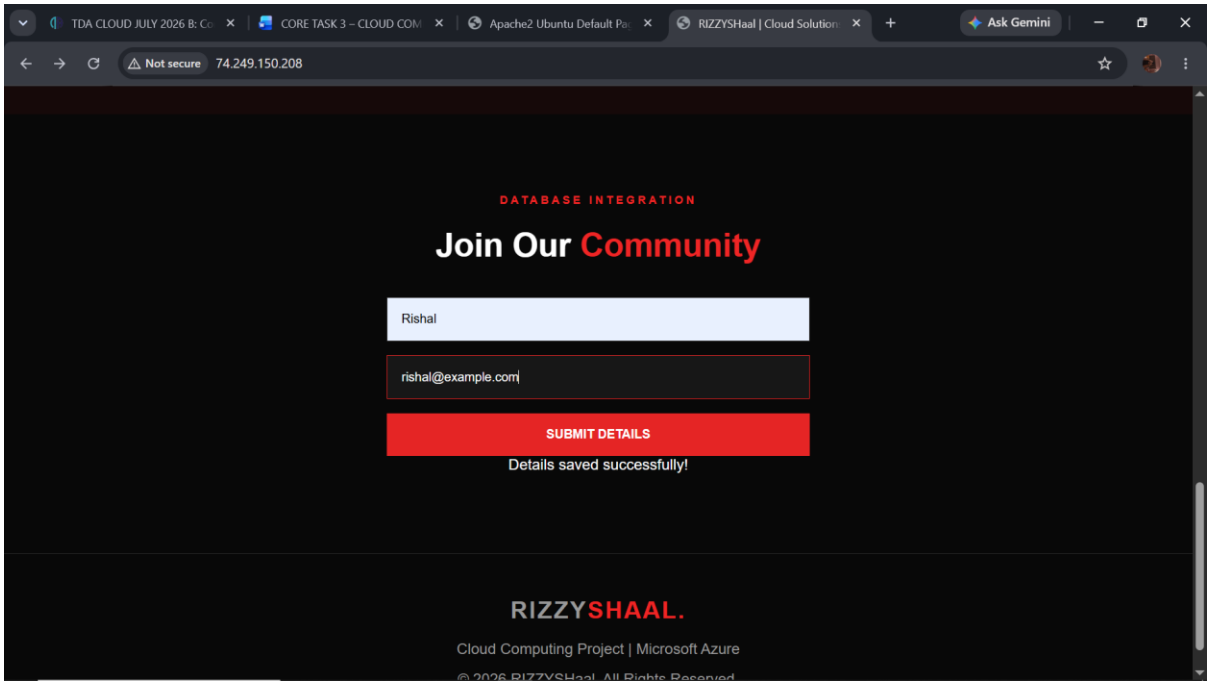
Verify that the database records are displayed.

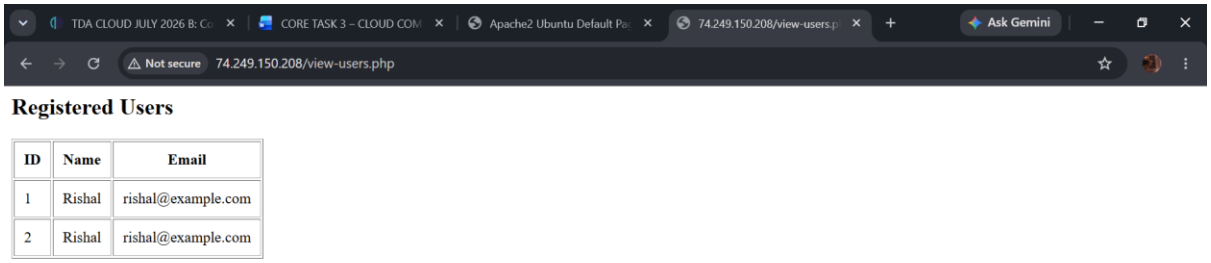
Expected sample record:

ID	Name	Email
1	Rishal	rishal@example.com

Explanation:

This test confirms that the PHP application can connect to Azure MySQL and retrieve information from the database.





Step 22: Test Storage Access

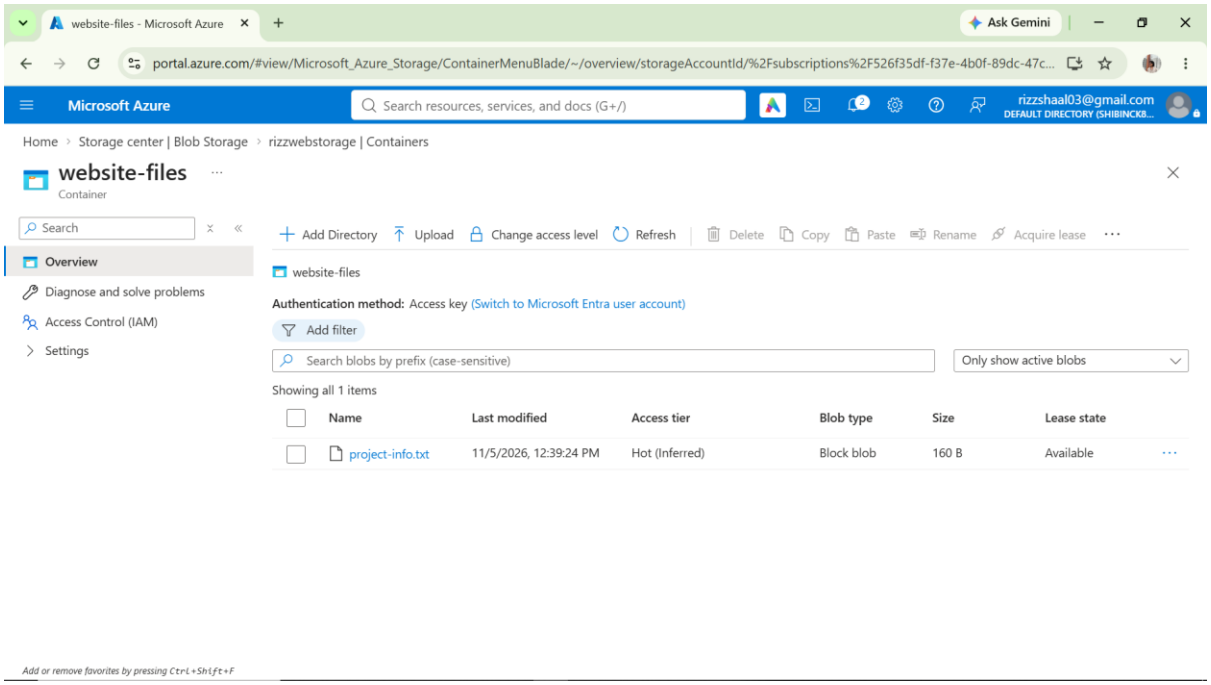
Console path: Azure Portal → Storage Accounts → Containers → `website-files`.

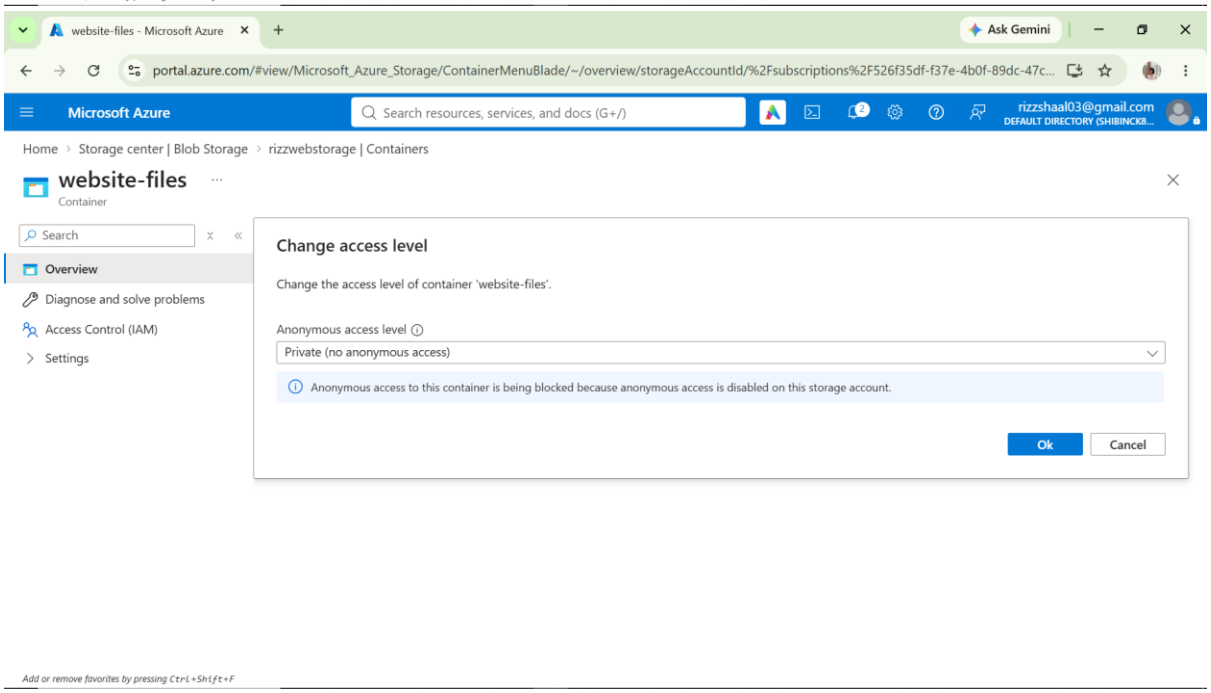
Verify:

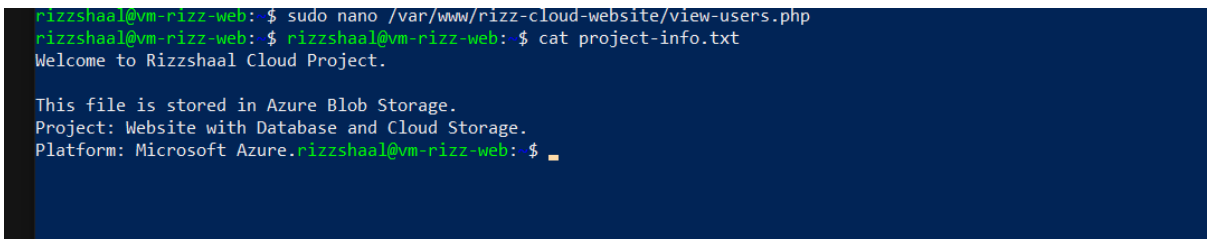
1. `project-info.txt` exists.
2. The container is private.
3. The downloaded file contents match the uploaded file.

Explanation:

This confirms that Azure Blob Storage can store files and that the VM can retrieve them securely.







Step 23: Configure Security

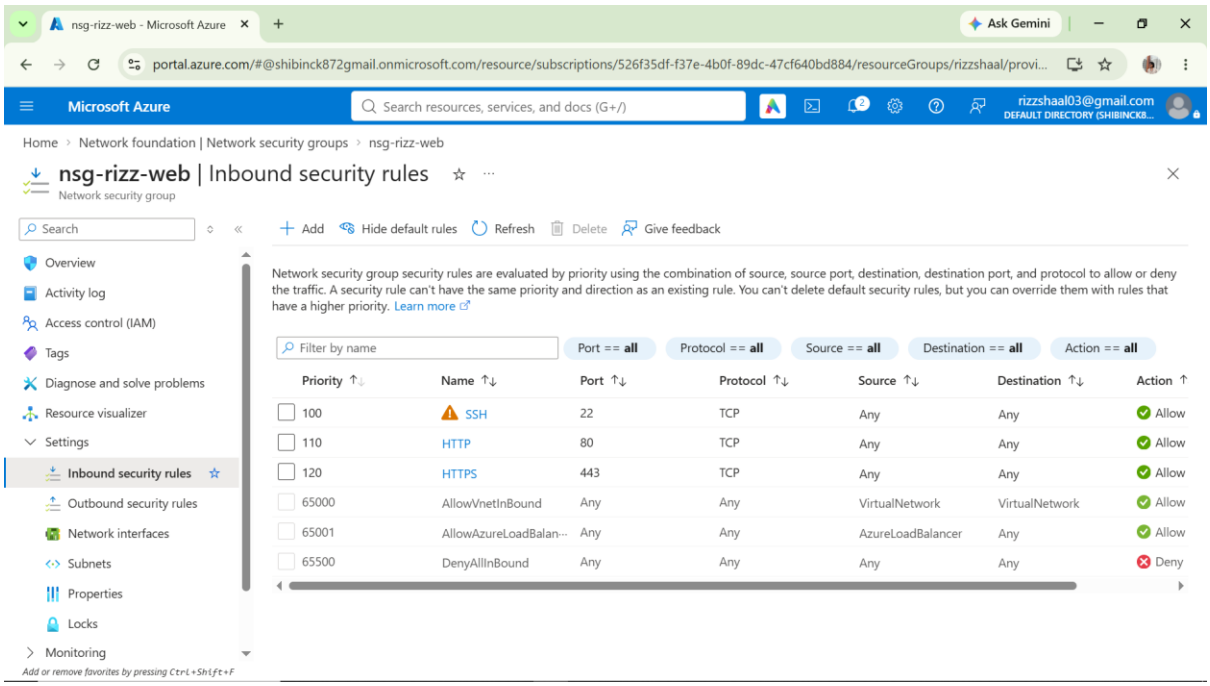
Console path: Azure Portal → NSG → Inbound Security Rules.

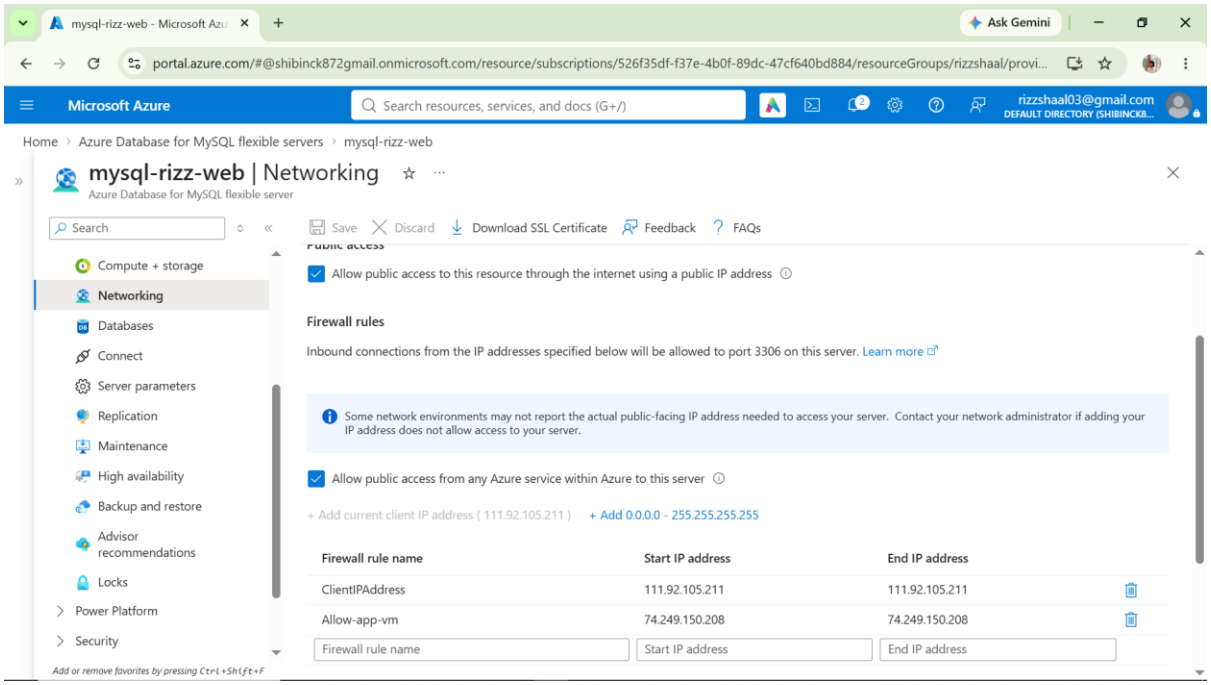
Security configuration:

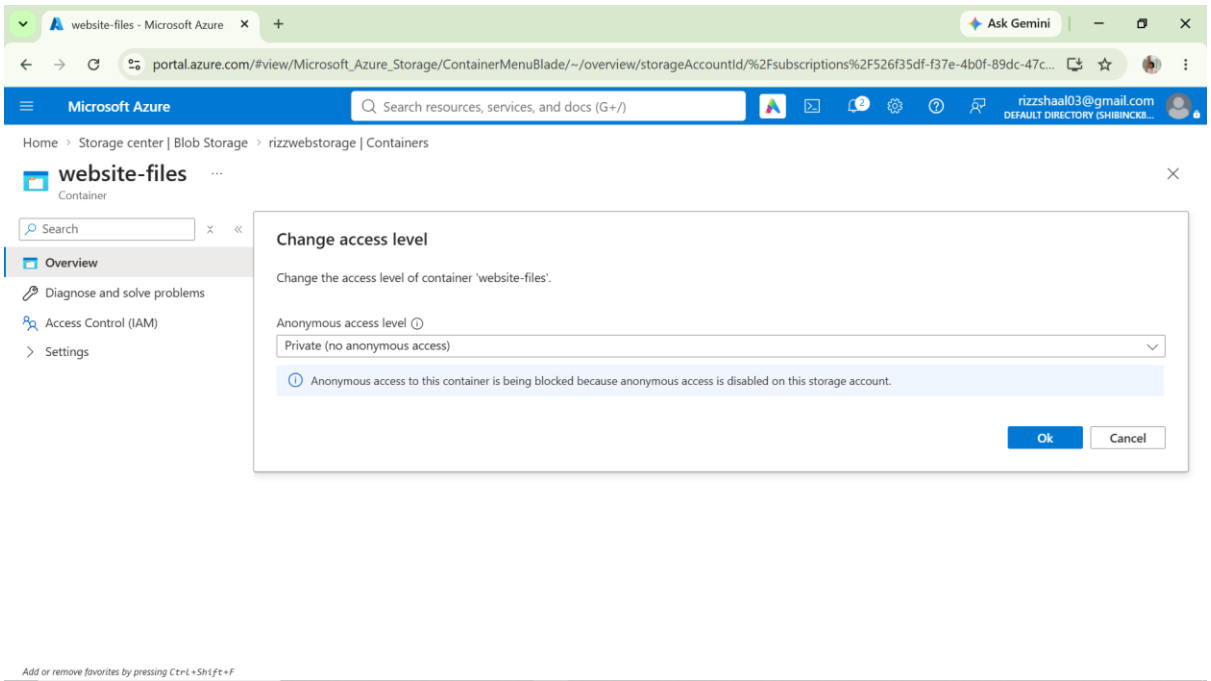
- SSH access restricted to the required IP address.
- HTTP access enabled for website testing.
- HTTPS access allowed where configured.
- MySQL firewall restricted to the required VM source.
- Blob Container configured as private.
- Managed Identity used for storage authentication.
- Database credentials stored in a protected configuration file.

Explanation:

Security configurations protect the application, database and storage from unauthorized access.








```
Project: Website with Database and Cloud Storage.
Platform: Microsoft Azure.rizzshaal@vm-rizz-web: $ ls -l /etc/rizz-cloud-db.conf
-rw-r----- 1 root www-data 106 Nov  5 11:39 /etc/rizz-cloud-db.conf
rizzshaal@vm-rizz-web: $
```

Step 24: Monitoring

Console path: Azure Portal → Virtual Machines → `vm-rizz-web` → Monitoring.

Check:

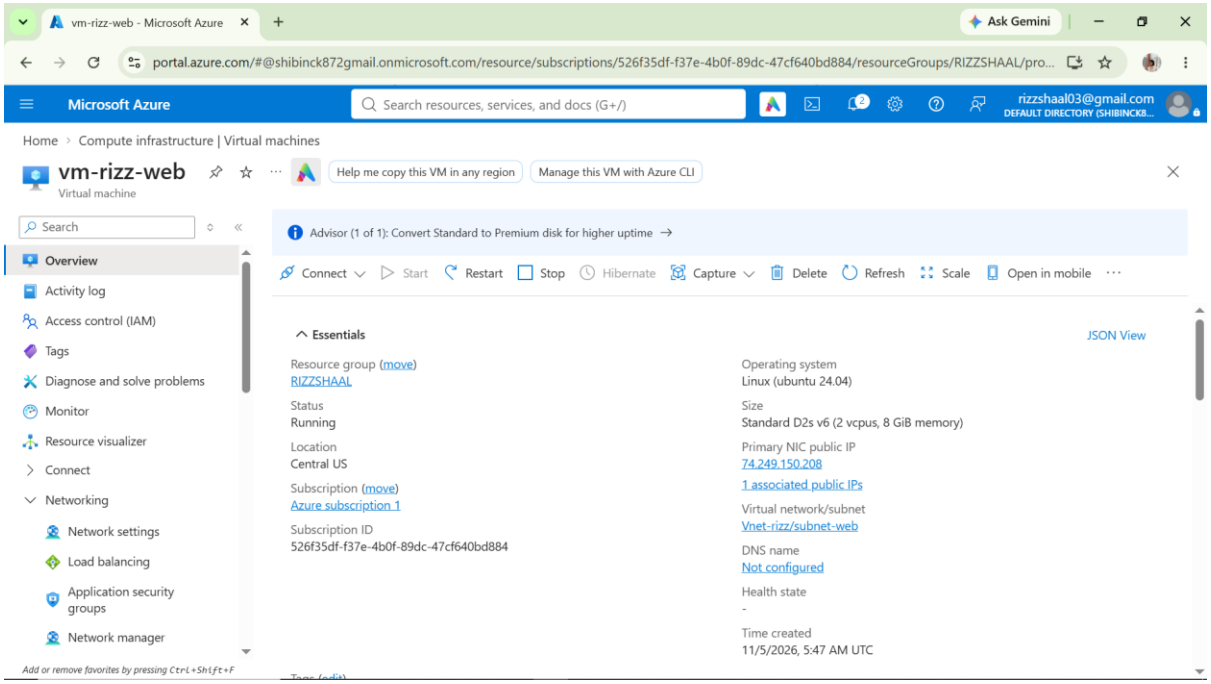
- VM running status.
- CPU utilization.
- Network activity.
- Disk usage.
- Boot diagnostics.

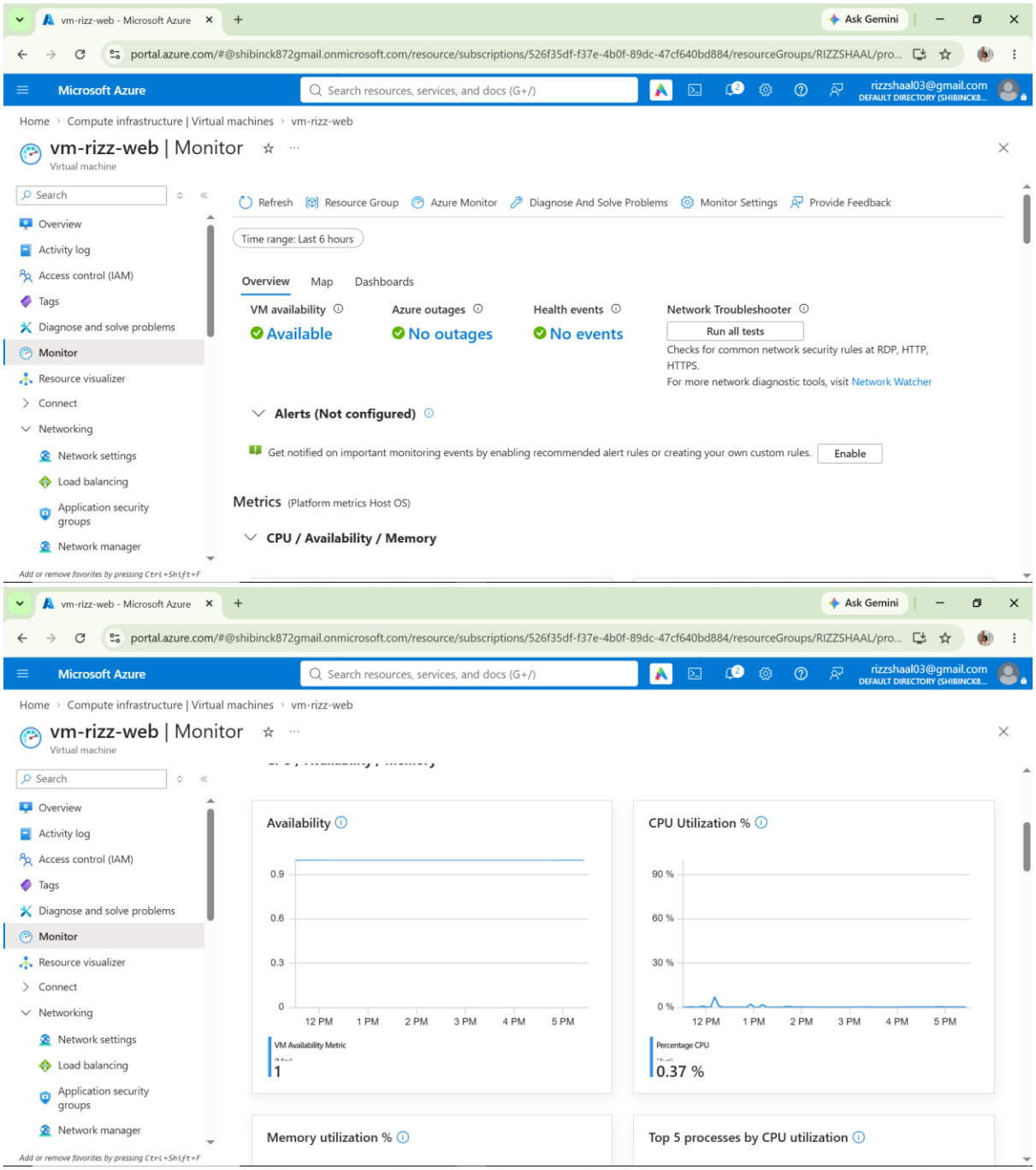
For database monitoring:

Azure Portal → MySQL Flexible Server → Monitoring.

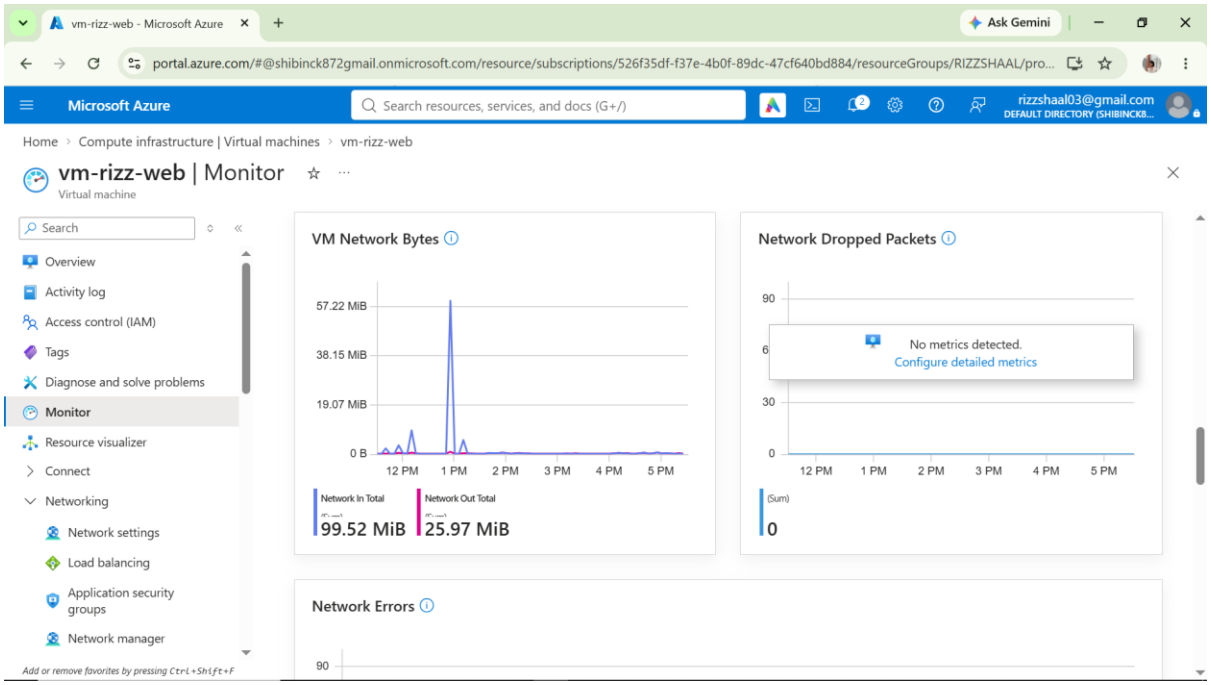
Explanation:

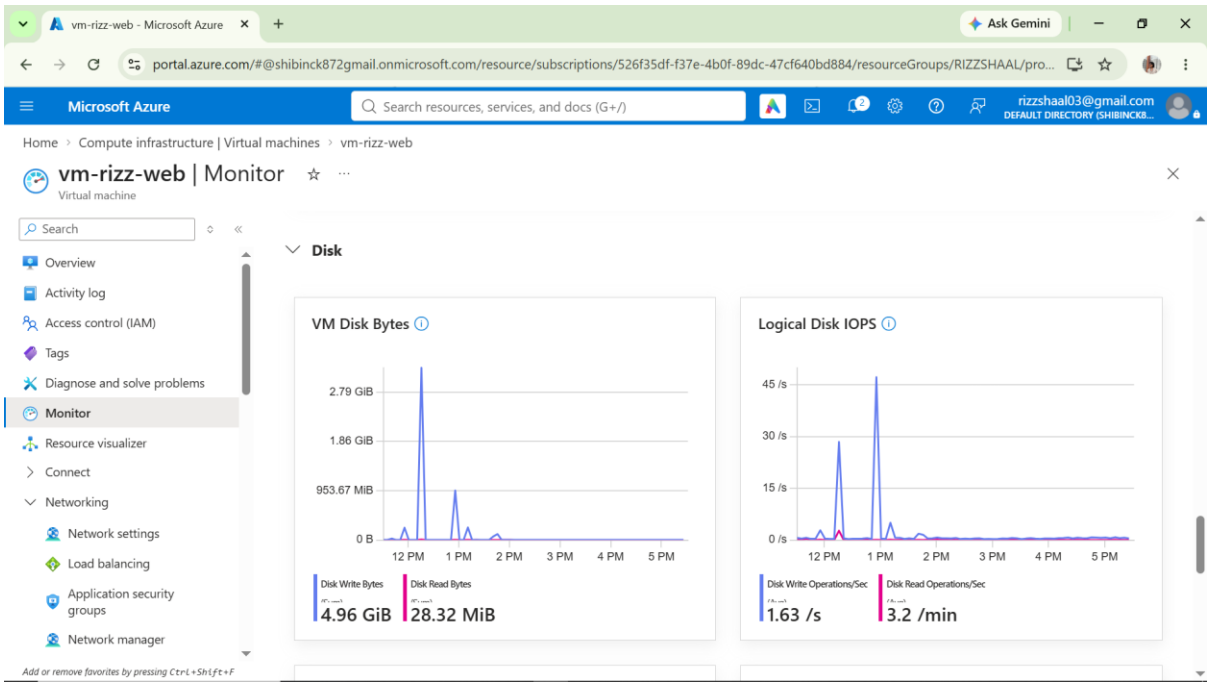
Monitoring helps identify resource usage, performance issues and possible failures.




vm-rizz-web - Microsoft Azure
Ask Gemini
portal.azure.com/#@shibinck872gmail.onmicrosoft.com/resource/subscriptions/526f35df-f37e-4b0f-89dc-47cf640bd884/resourceGroups/RIZZSHAAL/pro...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Compute infrastructure | Virtual machines › vm-rizz-web
vm-rizz-web | Monitor
Virtual machine
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Network settings
Load balancing
Application security groups
Network manager
Add or remove favorites by pressing Ctrl+Shift+F
Refresh
Resource Group
Azure Monitor
Diagnose And Solve Problems
Monitor Settings
Provide Feedback
Time range: Last 6 hours
Overview
Map
Dashboards
VM availability
Available
Azure outages
No outages
Health events
No events
Network Troubleshooter
Run all tests
Checks for common network security rules at RDP, HTTP, HTTPS.
For more network diagnostic tools, visit Network Watcher
Alerts (Not configured)
Get notified on important monitoring events by enabling recommended alert rules or creating your own custom rules.
Enable
Metrics (Platform metrics Host OS)
CPU / Availability / Memory

vm-rizz-web - Microsoft Azure
Ask Gemini
portal.azure.com/#@shibinck872gmail.onmicrosoft.com/resource/subscriptions/526f35df-f37e-4b0f-89dc-47cf640bd884/resourceGroups/RIZZSHAAL/pro...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Compute infrastructure | Virtual machines › vm-rizz-web
vm-rizz-web | Monitor
Virtual machine
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Network settings
Load balancing
Application security groups
Network manager
Add or remove favorites by pressing Ctrl+Shift+F
Availability
0.9
0.6
0.3
0
12 PM
1 PM
2 PM
3 PM
4 PM
5 PM
VM Availability Metric
1
CPU Utilization %
90 %
60 %
30 %
0 %
12 PM
1 PM
2 PM
3 PM
4 PM
5 PM
Percentage CPU
0.37 %
Memory utilization %
Top 5 processes by CPU utilization


vm-rizz-web - Microsoft Azure
Ask Gemini
portal.azure.com/#@shibinck872gmail.onmicrosoft.com/resource/subscriptions/526f35df-f37e-4b0f-89dc-47cf640bd884/resourceGroups/RIZZSHAAL/pro...
Microsoft Azure
Search resources, services, and docs (G+/)
rizzshaal03@gmail.com
DEFAULT DIRECTORY (SHIBINCK8...
Home › Compute infrastructure | Virtual machines › vm-rizz-web
vm-rizz-web | Monitor
Virtual machine
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Network settings
Load balancing
Application security groups
Network manager
Add or remove favorites by pressing Ctrl+Shift+F
VM Network Bytes
57.22 MiB
38.15 MiB
19.07 MiB
0 B
12 PM
1 PM
2 PM
3 PM
4 PM
5 PM
Network In Total
99.52 MiB
Network Out Total
25.97 MiB
Network Dropped Packets
90
6
30
0
No metrics detected.
Configure detailed metrics
(Sum)
0
Network Errors
90


vm-rizz-web - Microsoft Azure
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Disk
VM Disk Bytes
2.79 GiB
1.86 GiB
953.67 MiB
0 B
12 PM
1 PM
2 PM
3 PM
4 PM
5 PM
Disk Write Bytes
4.96 GiB
Disk Read Bytes
28.32 MiB
Logical Disk IOPS
45 /s
30 /s
15 /s
0 /s
12 PM
1 PM
2 PM
3 PM
4 PM
5 PM
Disk Write Operations/Sec
1.63 /s
Disk Read Operations/Sec
3.2 /min

10. Conclusion

This practical provided hands-on experience in deploying a cloud-based website using Microsoft Azure.

I learned how to create and configure Azure networking resources, deploy a Linux Virtual Machine, install Apache, create an Azure MySQL database and integrate cloud storage using Azure Blob Storage.

I also learned how to transfer website source code using ZIP and SCP, configure database connectivity, manage access permissions, and troubleshoot common cloud deployment issues.

Overall, this project improved my practical knowledge of cloud computing, Linux administration, networking, database management and cloud storage integration.